CSE 150A-250A AI: Probabilistic Models

Lecture 6

Fall 2026

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Slides adapted from previous versions of the course (Prof. Lawrence, Prof. Alvarado, Prof Berg-Kirkpatrick)

Agenda

Polytrees

- Node clustering
- Cutset conditioning

Approximate inference in loopy BNs

- Rejection sampling
- Likelihood weighting

Polytrees

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- Definition

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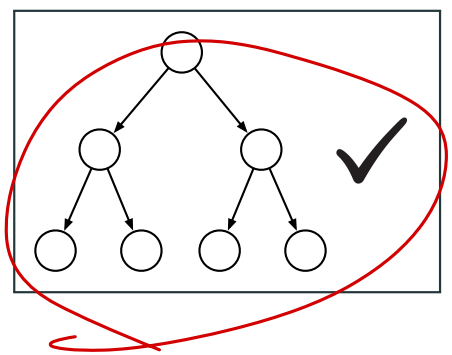
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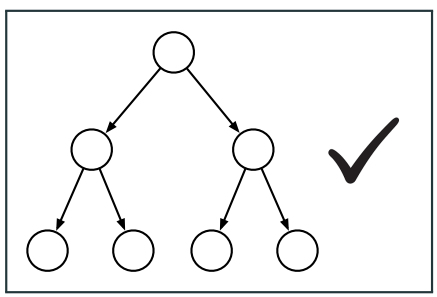
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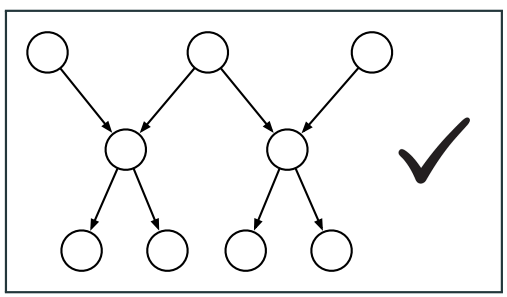
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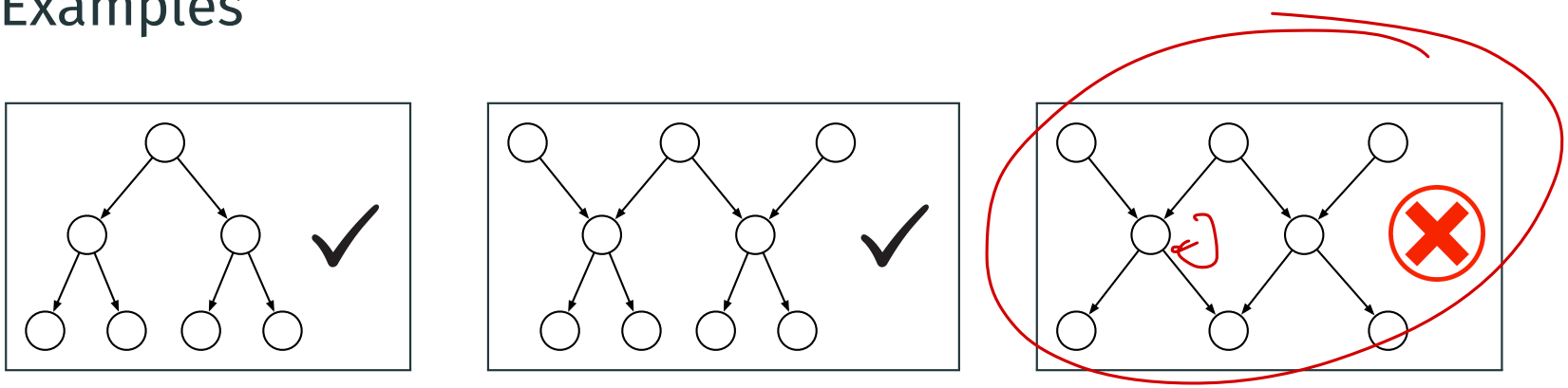
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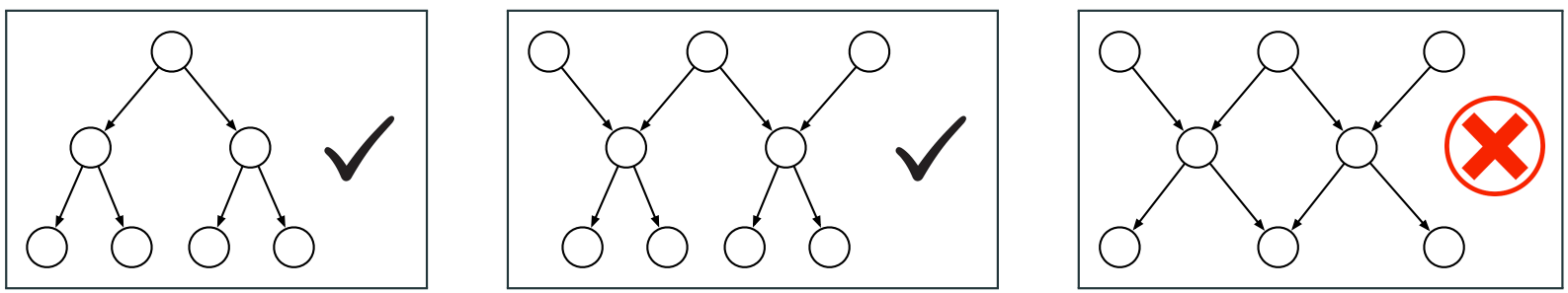
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- **Examples**



All trees are polytrees.

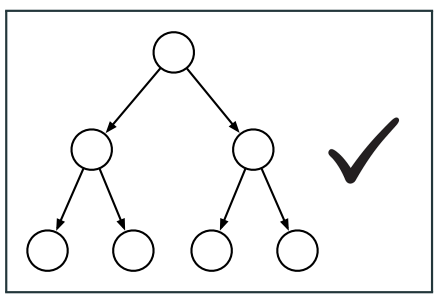
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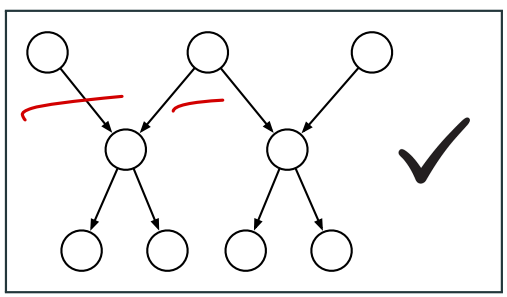
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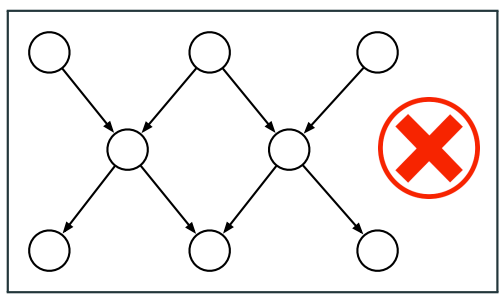


All trees are polytrees.



But not vice versa!

A node in a polytree may have multiple parents.



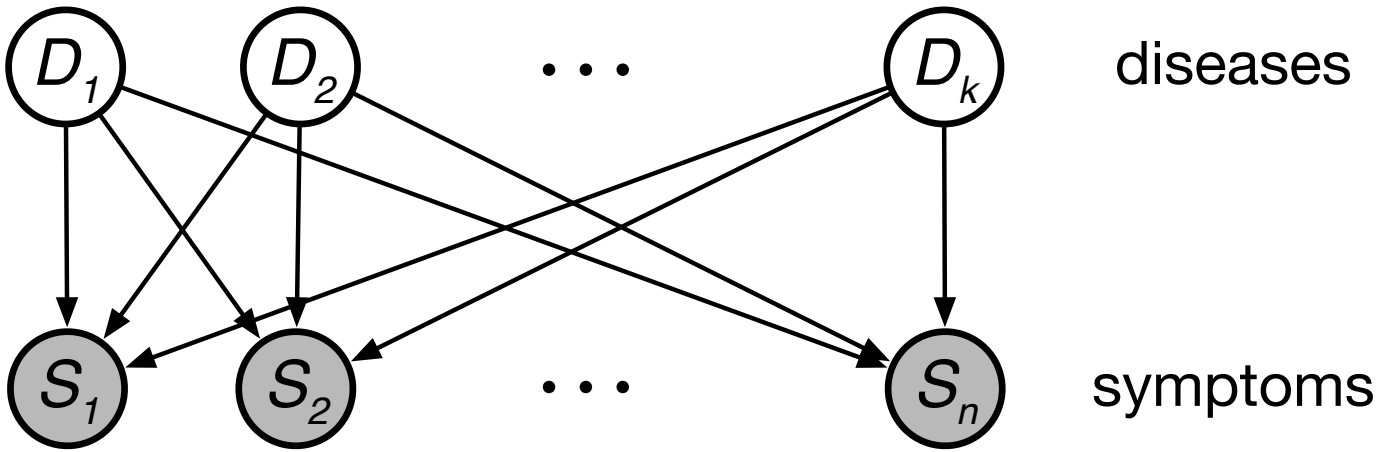
Exact Inference in loopy BNs

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But many interesting BNs are not polytrees!

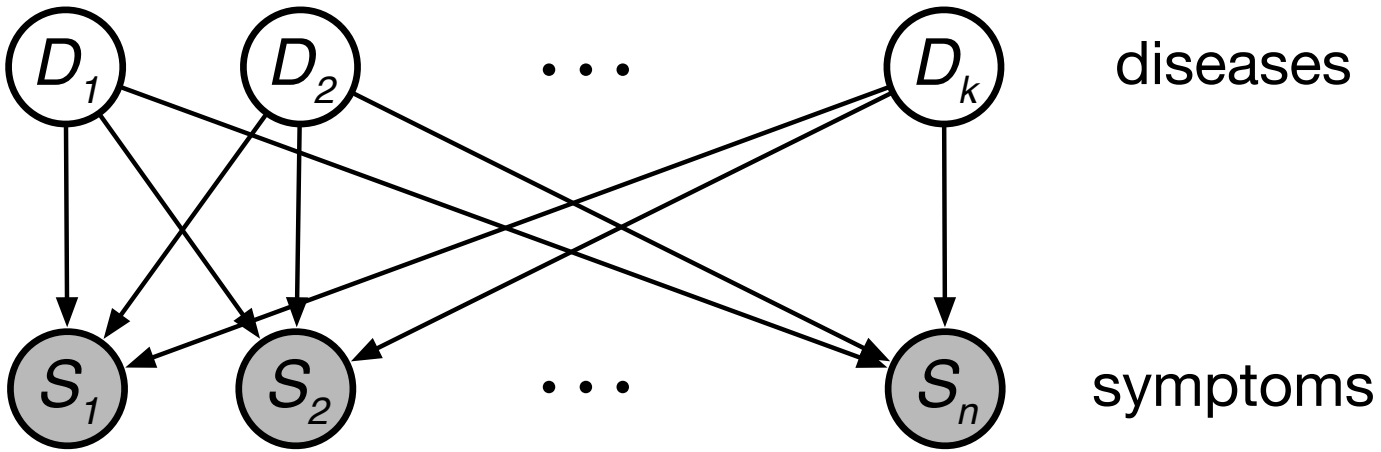
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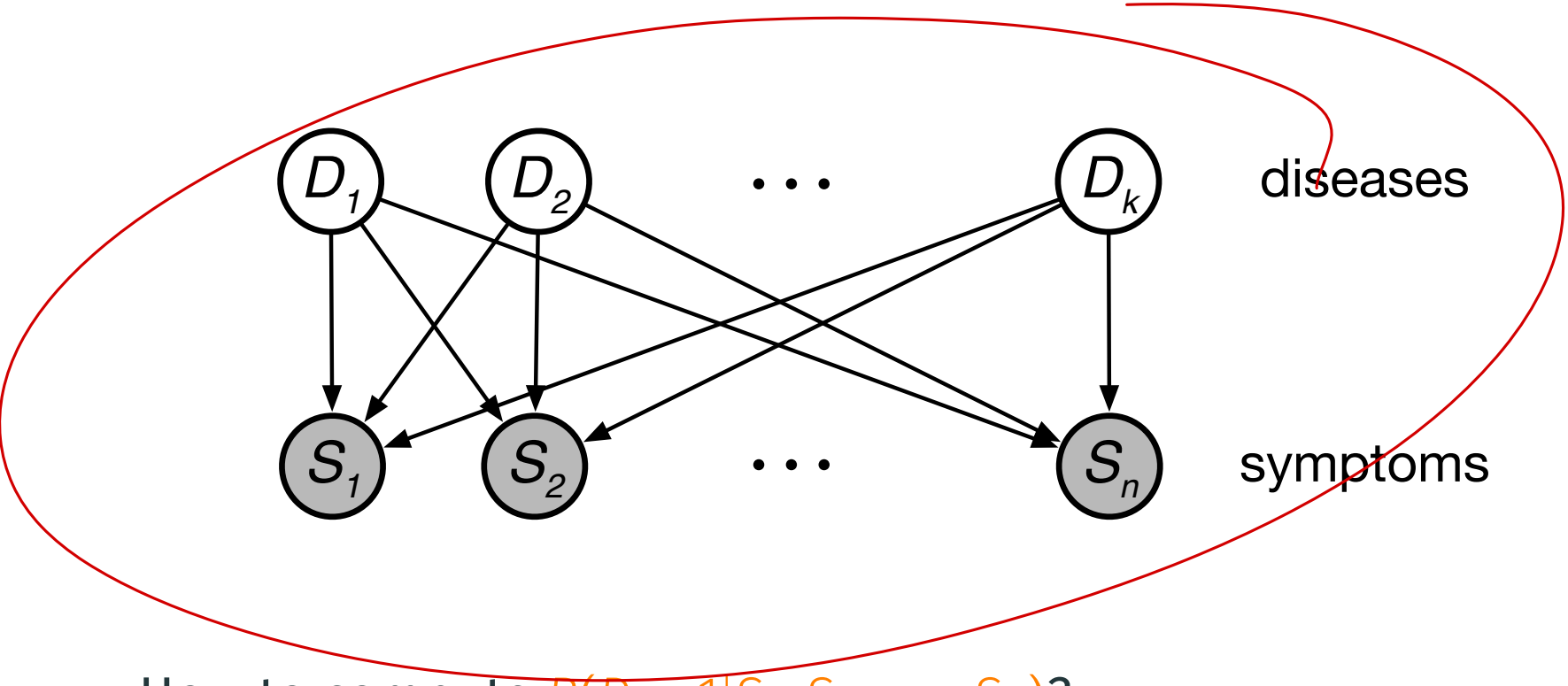
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How to compute $P(D_i = 1 | S_1, S_2, \ldots, S_n)$?

Exact Inference in loopy BNs

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How to compute $P(D_i = 1 | S_1, S_2, \ldots, S_n)$?

What are general strategies for inference in these BNs?

Exact inference in loopy BNs

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- Main idea

Exact inference in loopy BNs

- **Main idea**

  Can we transform a loopy BN into a polytree?

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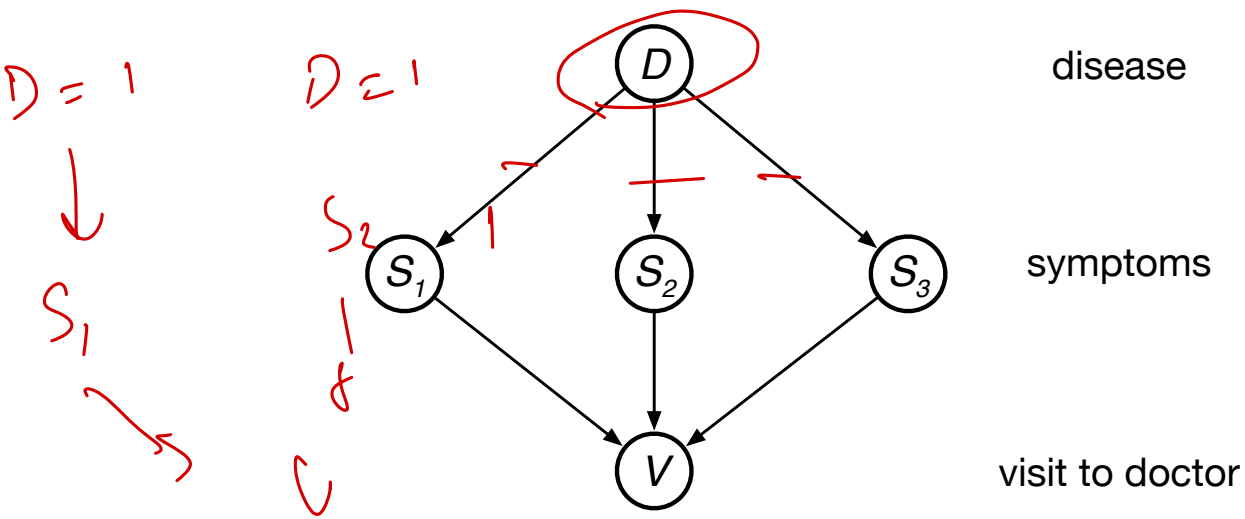
Exact inference in loopy BNs

- **Main idea**

  Can we transform a loopy BN into a polytree?
  If so, then we can run the exact inference algorithm.

- **Example**

  We'll use a simple BN with binary variables to illustrate two different ways of doing this.



1. Node clustering

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- **Key idea**

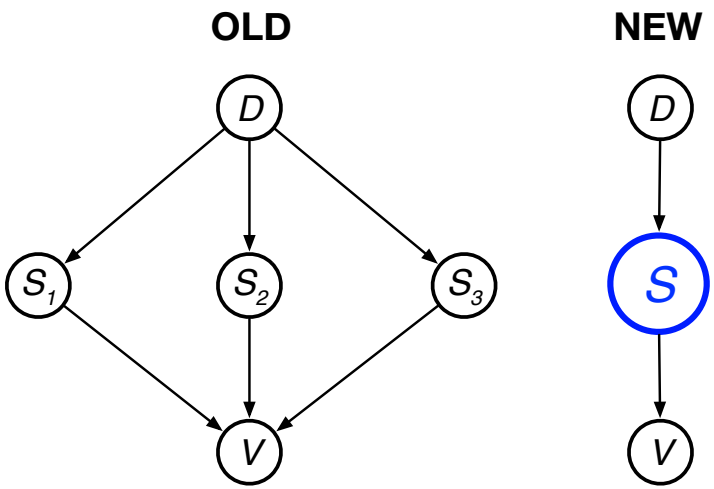
  Merge (well-chosen) nodes in the DAG to remove loops, so that what remains is a polytree.

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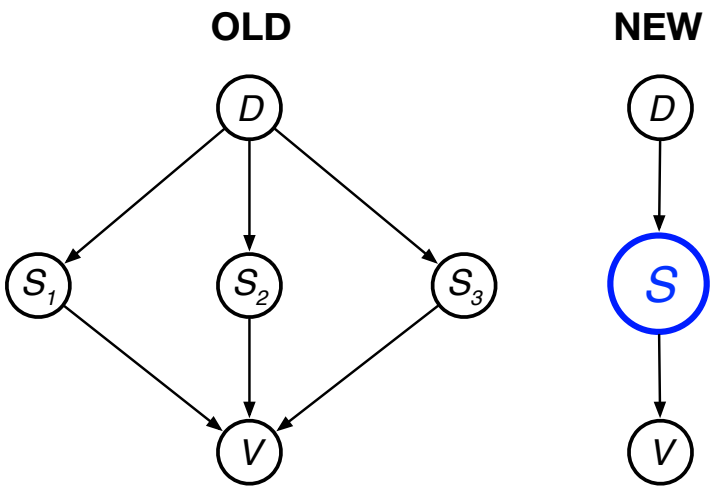


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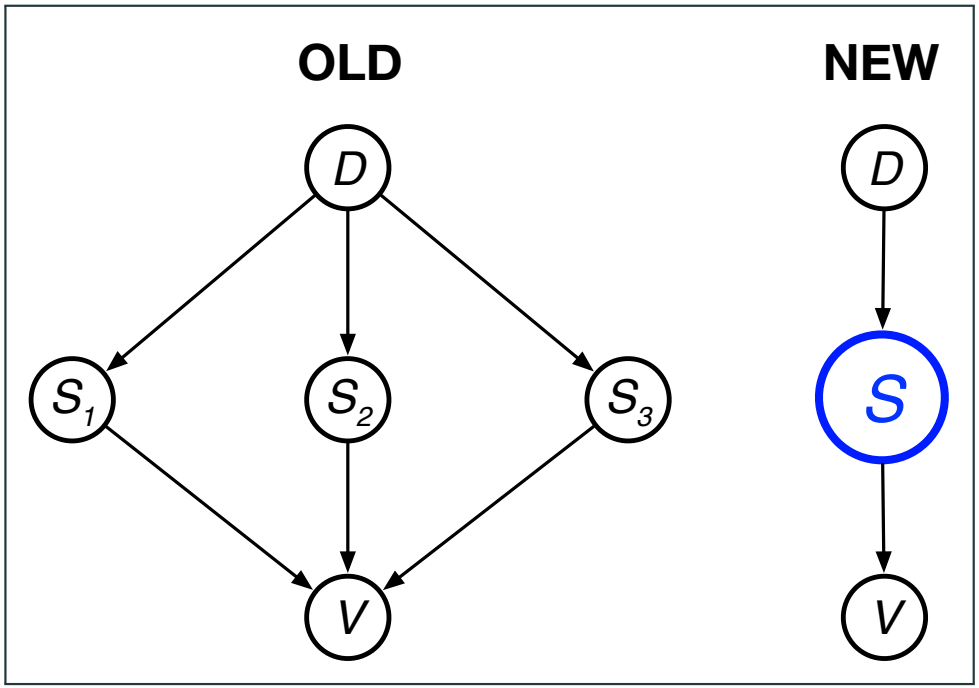
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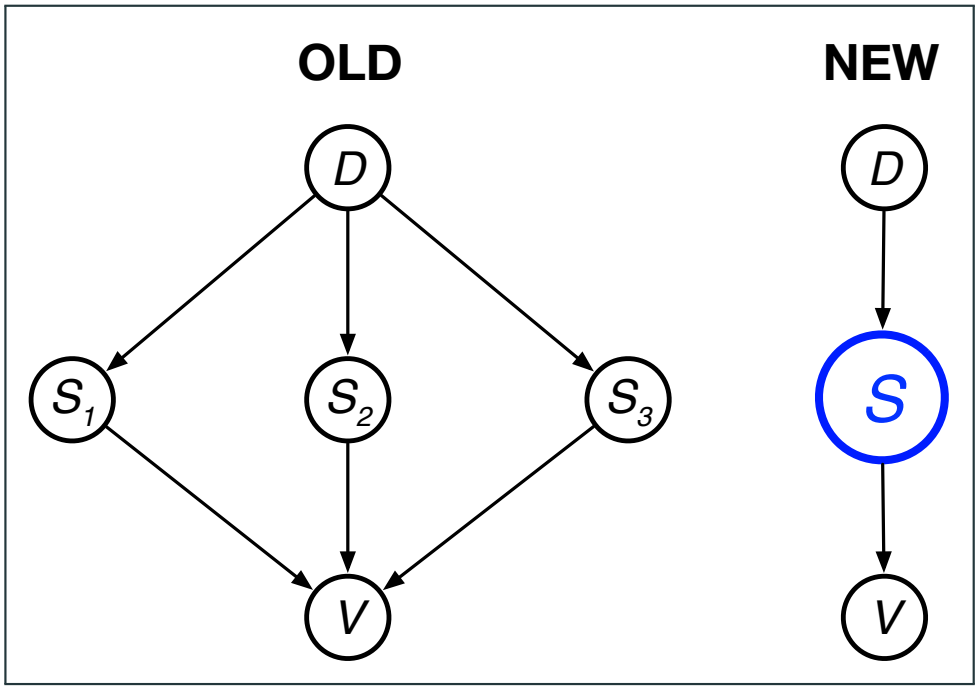
Cluster nodes $\{S_1, S_2, S_3\}$ into mega-node S.

Merge CPTs at these nodes into mega-CPT $P(S|D)$.

Old versus new nodes

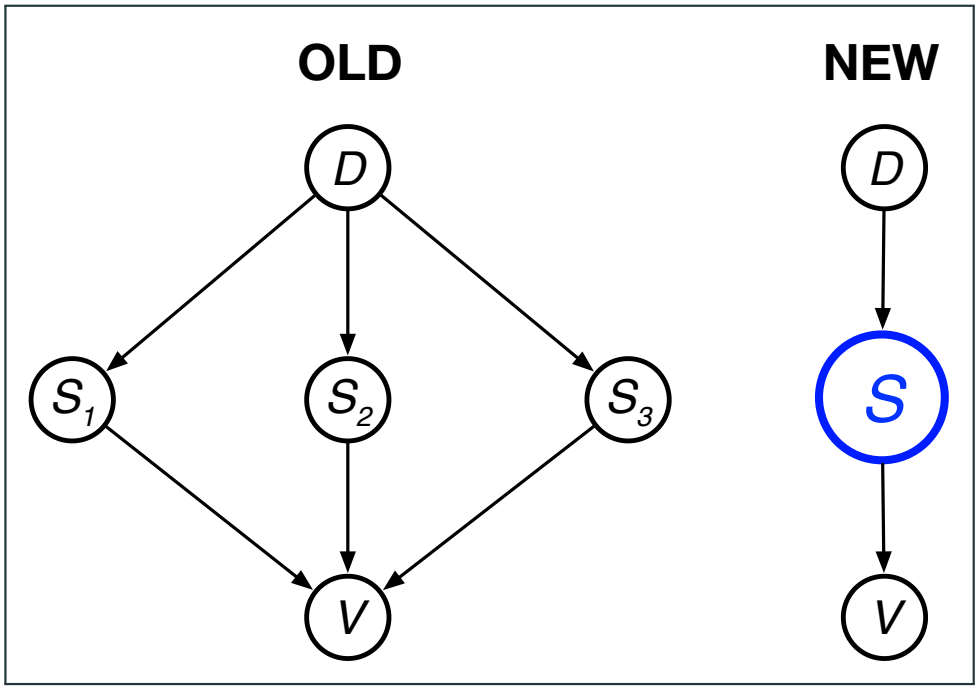


Old versus new nodes



S_3	S_2	S_1	S
0	0	0	0
0	0	1	1
0	1	0	2
0	1	1	3
1	0	0	4
1	0	1	5
1	1	0	6
1	1	1	7

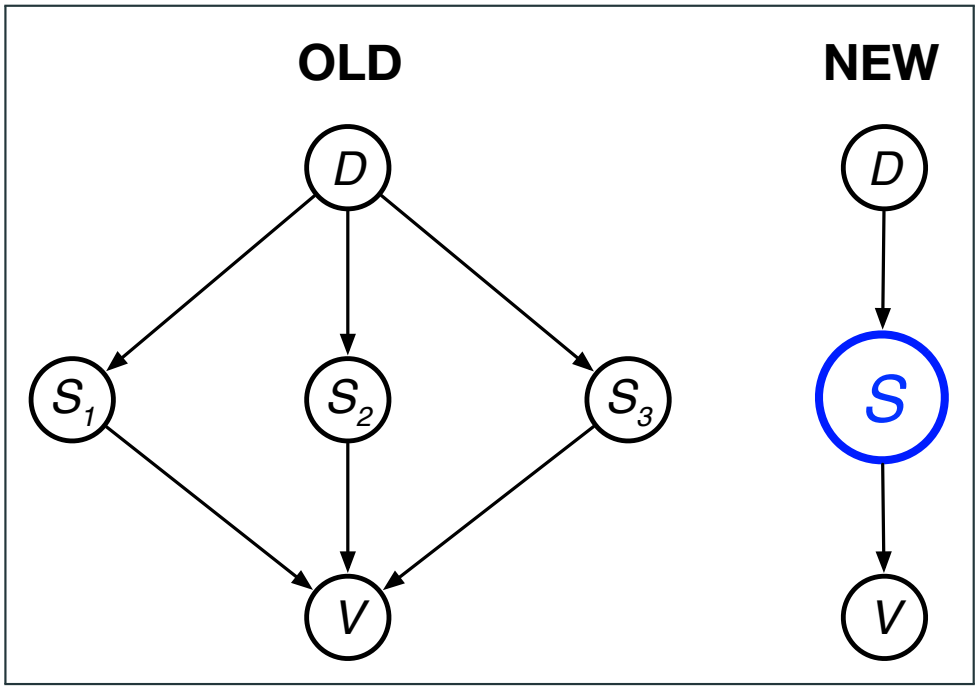
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Pro

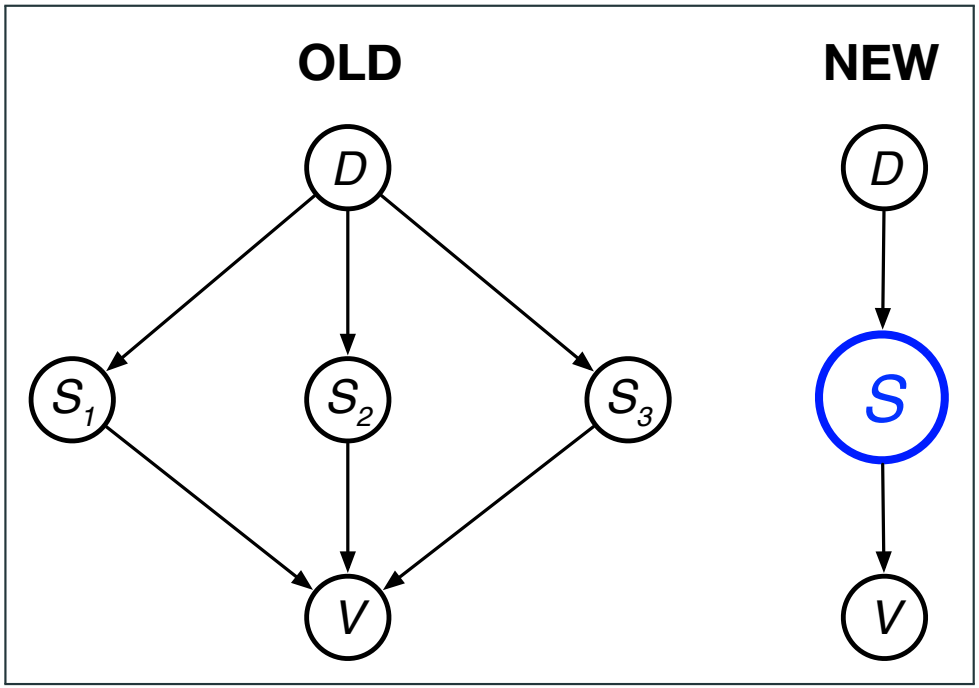
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Pro The graph simplifies to a polytree.

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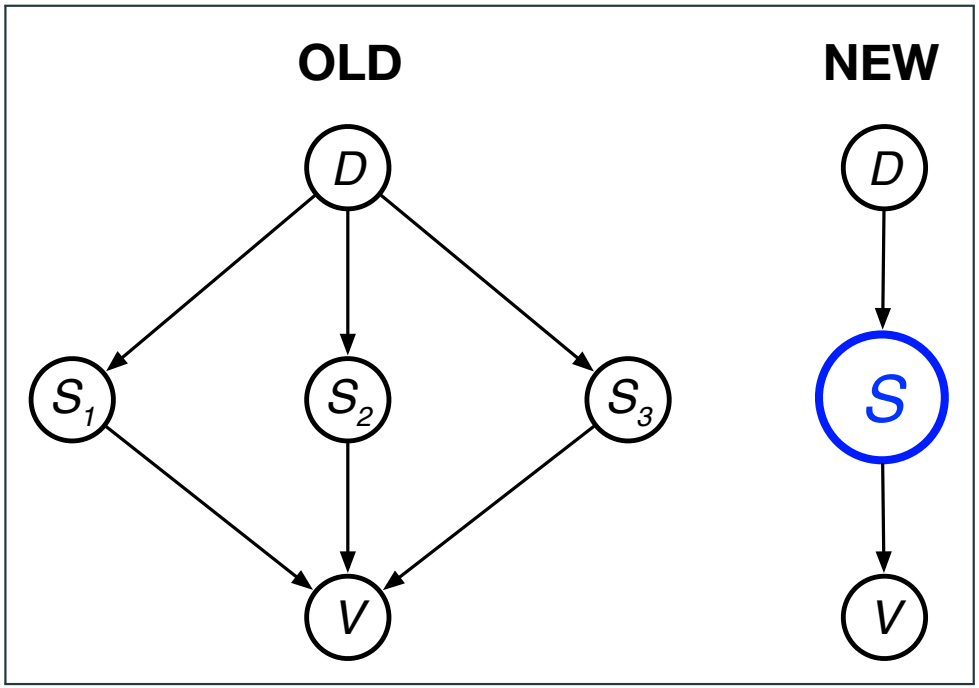


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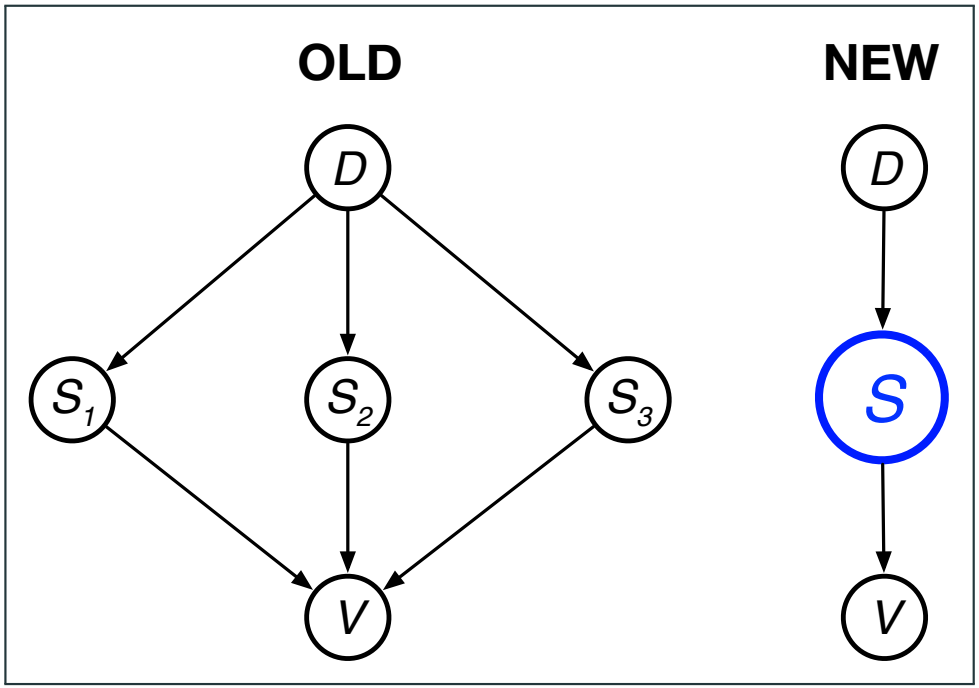


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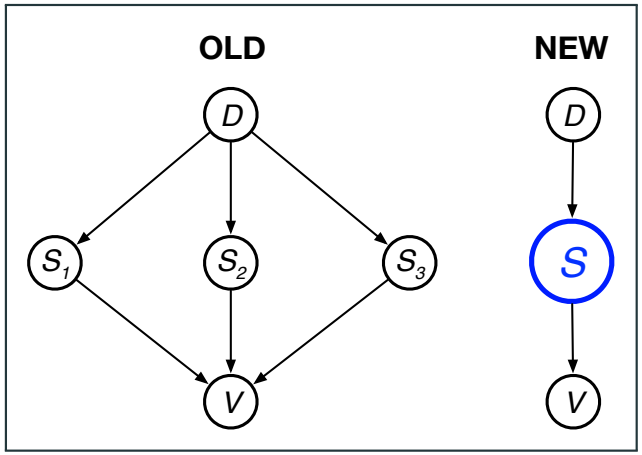
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$$|S| = |S_1| \cdot |S_2| \cdot |S_3| = 2^3 = 8$$

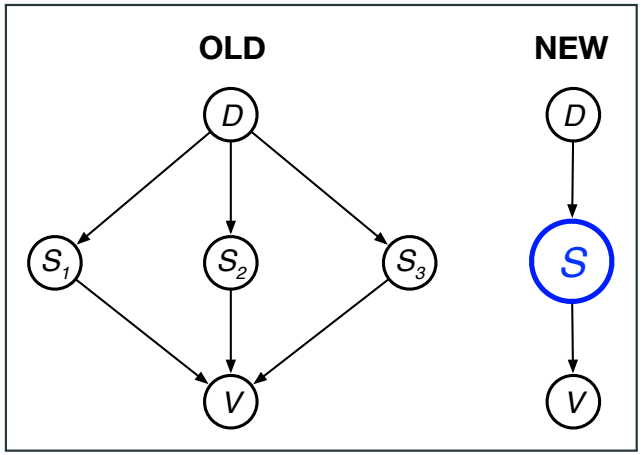
Old versus new CPTs

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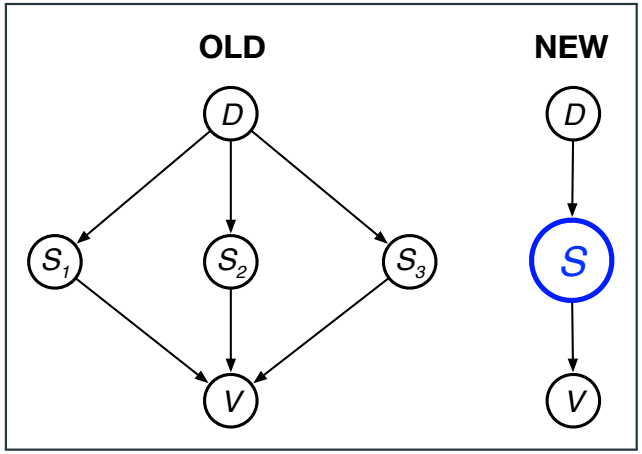
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OLD	NEW
$P(S_1\|D)$ $P(S_2\|D)$ $P(S_3\|D)$	$P(S\|D) = P(S_1, S_2, S_3\|D) = \prod_{i=1}^{3} P(S_i\|D)$
$P(V\|S_1, S_2, S_3)$	$P(V\|S) = P(V\|S_1, S_2, S_3)$

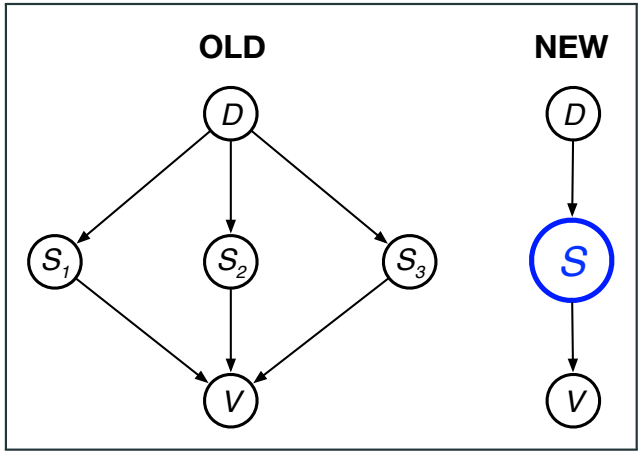
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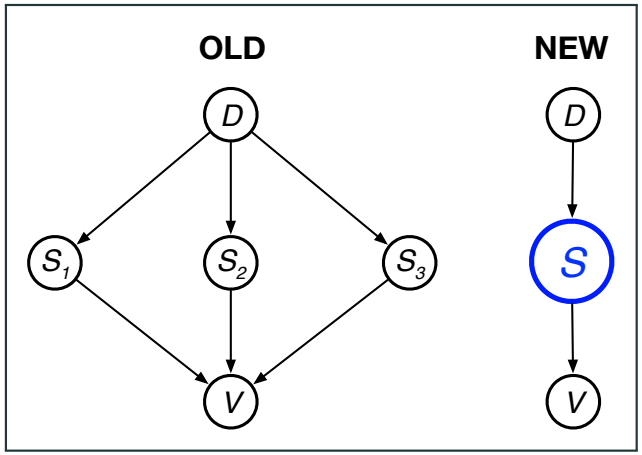
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To calculate the new CPTs:

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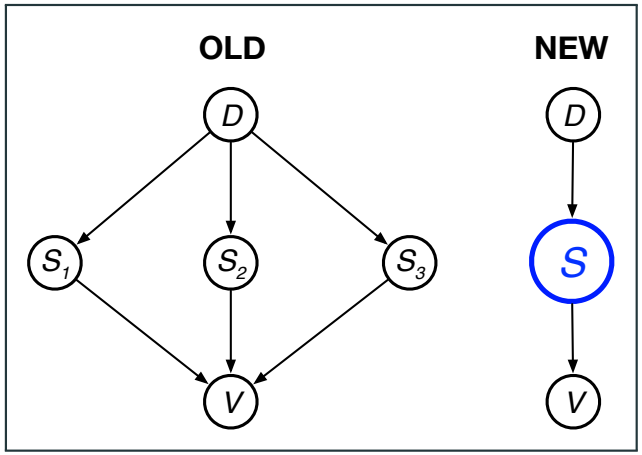


To calculate the new CPTs:

$$P(S=5|D=0)$$

Worked example

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0	0	0	0
0	0	1	1
0	1	0	2
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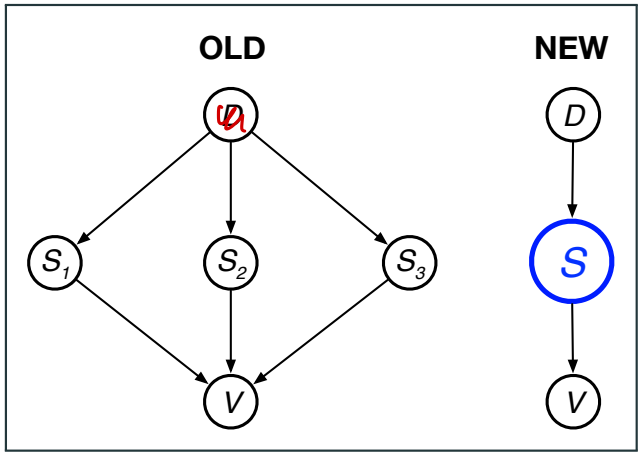


To calculate the new CPTs:

$$P(S=5|D=0) \quad = \quad P(S_1=1, S_2=0, S_3=1|D=0)$$

Worked example

S_3	S_2	S_1	S
0	0	0	0
0	0	1	1
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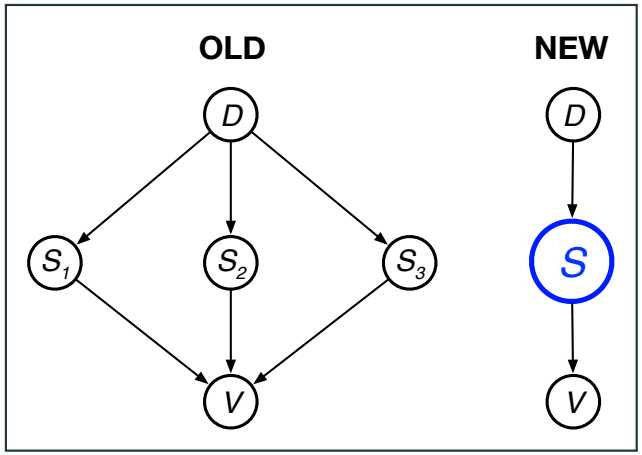


To calculate the new CPTs:

$$\begin{aligned} P(S=5|D=0) &= P(S_1=1, S_2=0, S_3=1|D=0) \\ &= P(S_1=1|D=0)\, P(S_2=0|D=0)\, P(S_3=1|D=0) \end{aligned}$$

Worked example

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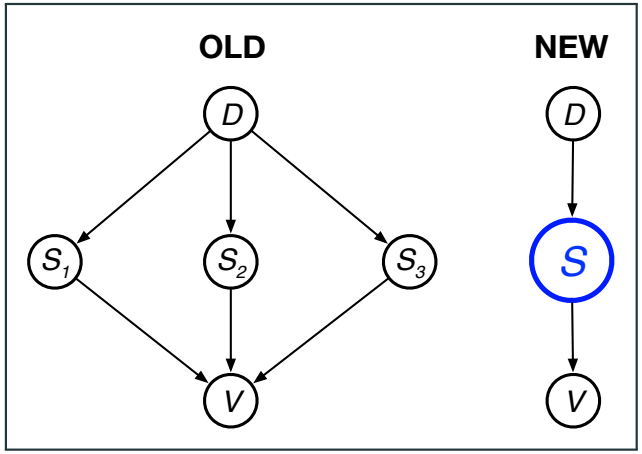
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$$P(V=1|S=5)$$

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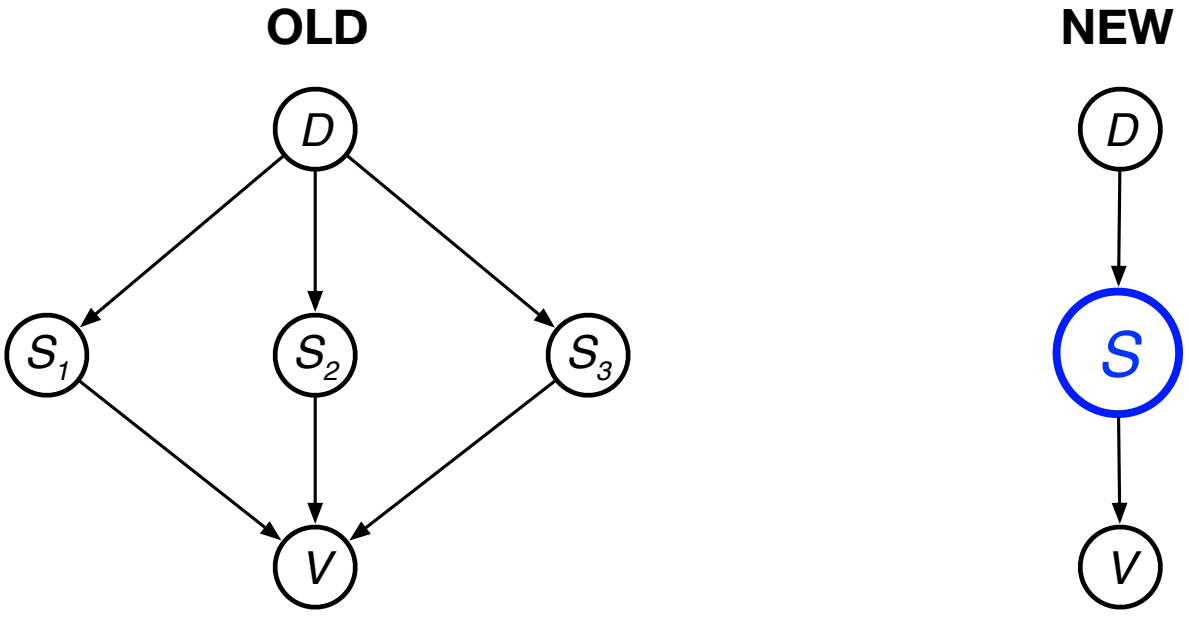


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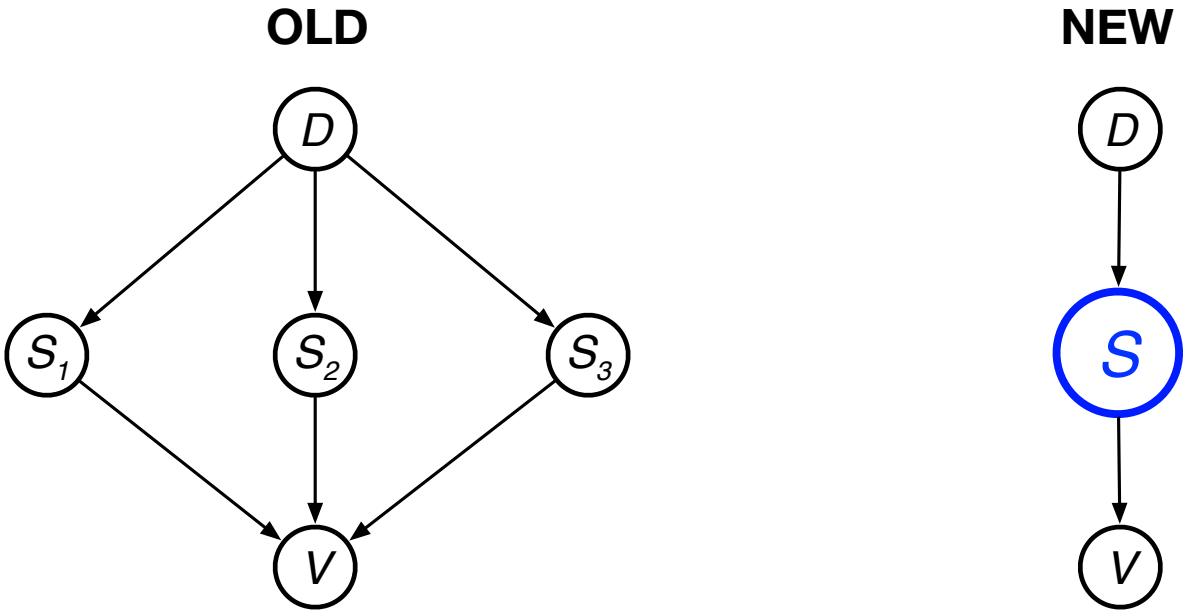
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$$P(V=1|S=5) = P(V|S_1=1, S_2=0, S_3=1)$$

Which nodes to cluster?

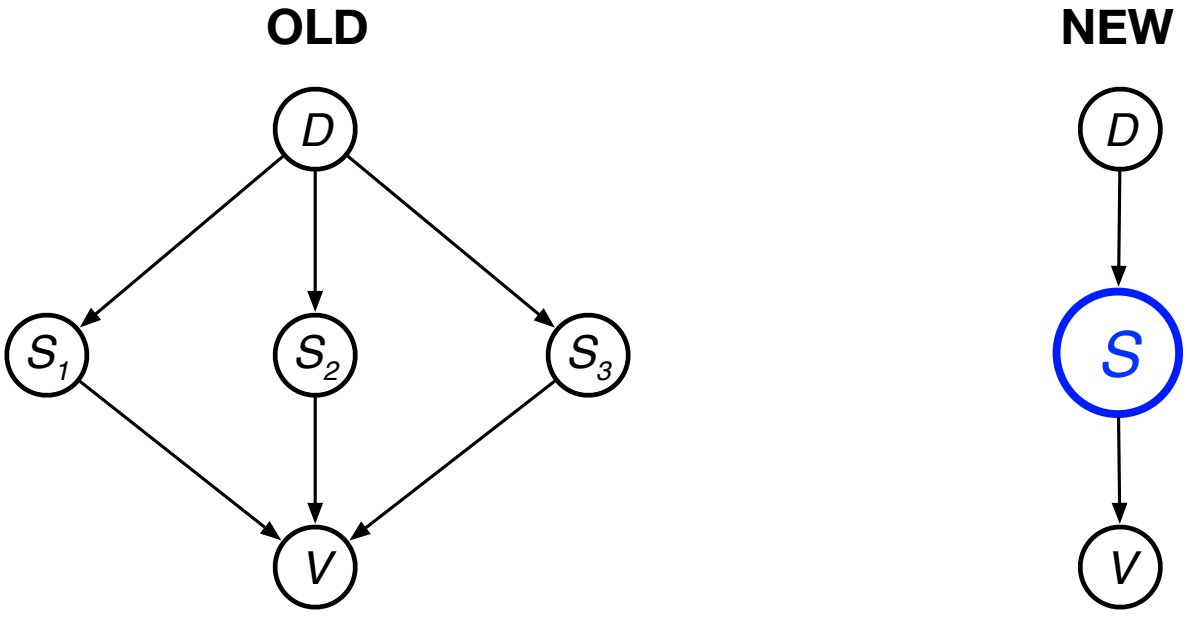


Which nodes to cluster?



In this BN, we can eyeball the right nodes to cluster.

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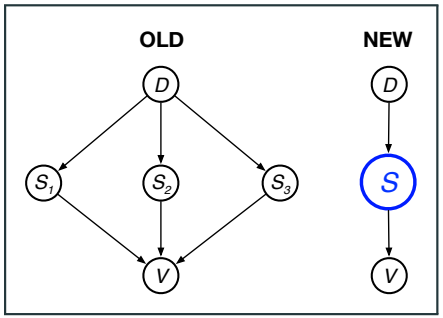
In this BN, we can eyeball the right nodes to cluster.

What about in larger BNs?

General case

General case

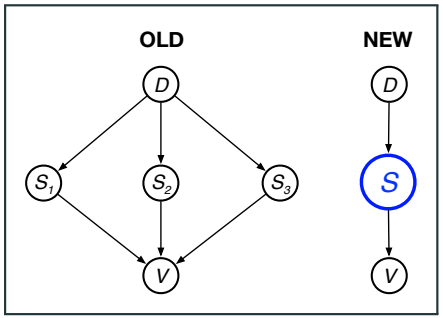
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General case

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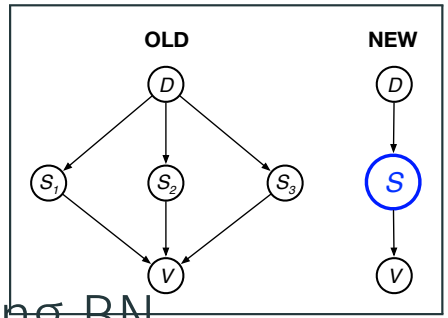
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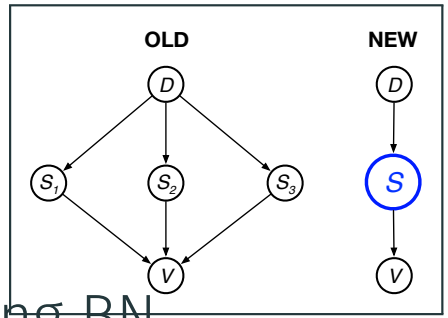
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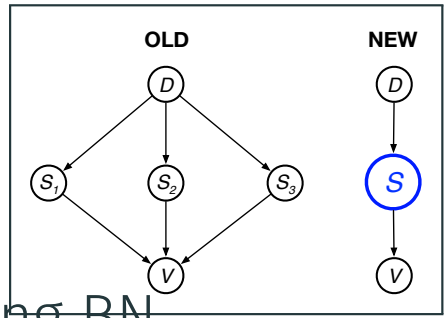


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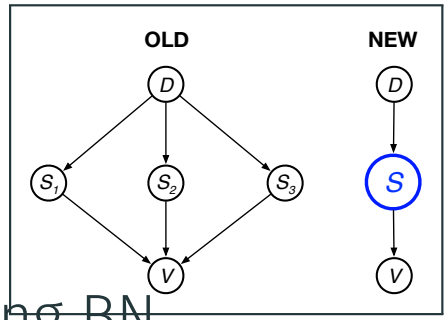
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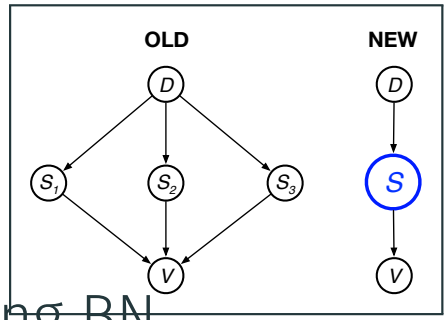
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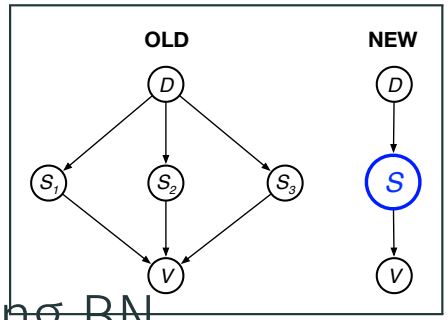
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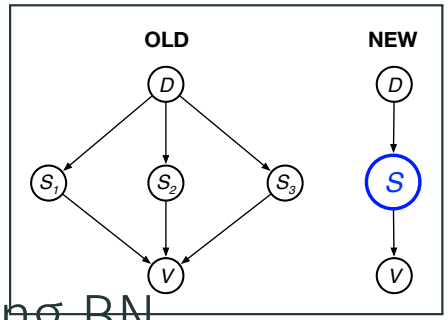
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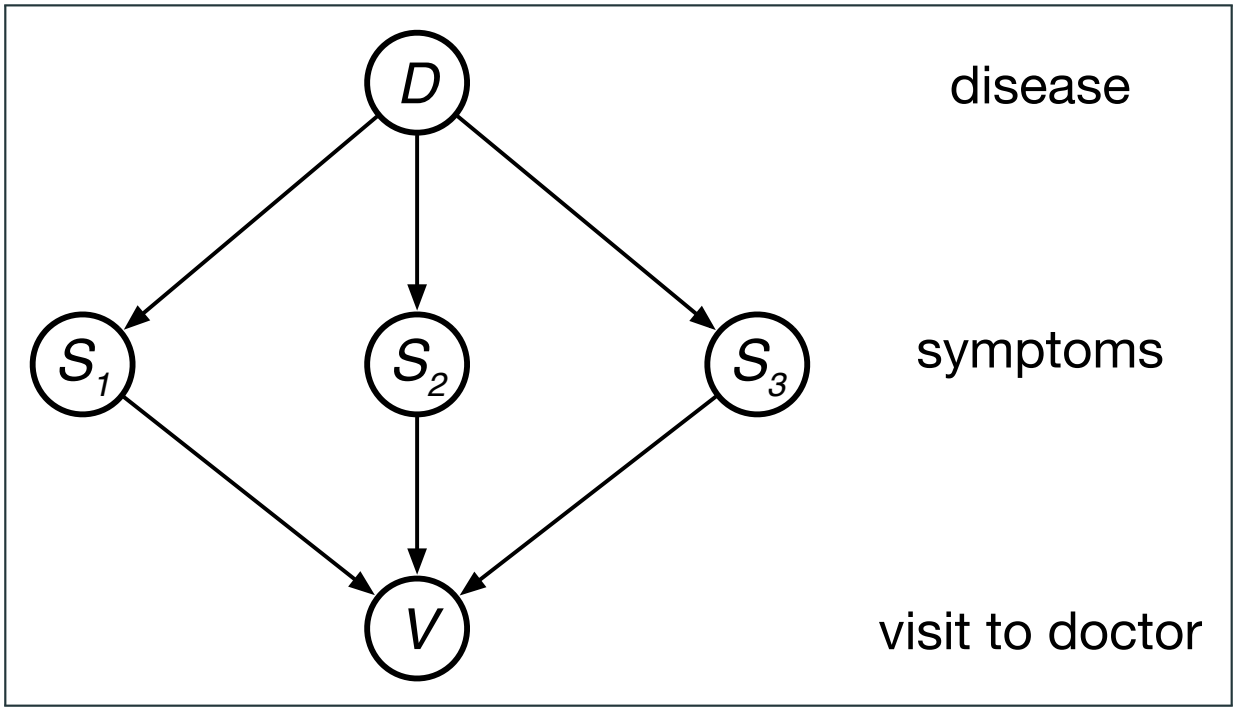
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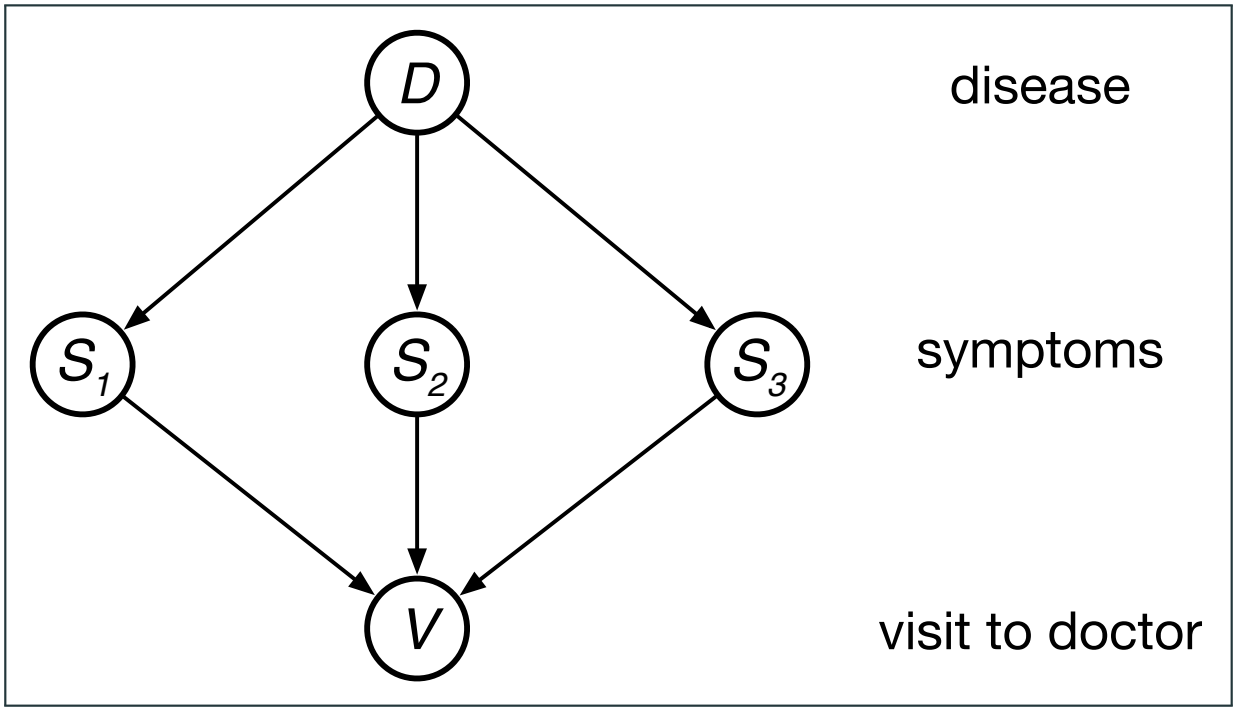
  Which clustering leads to maximally efficient inference?
  There is no efficient algorithm to find this!

A different approach?

A different approach?



A different approach?



What if, instead of merging nodes, we remove them?

2. Cutset conditioning

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- Key idea

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  Remove one or more nodes by instantiating them as evidence.

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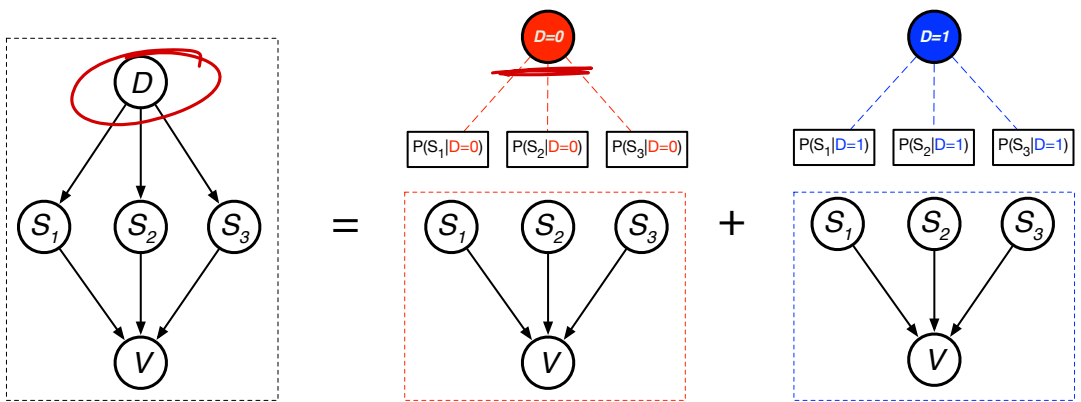
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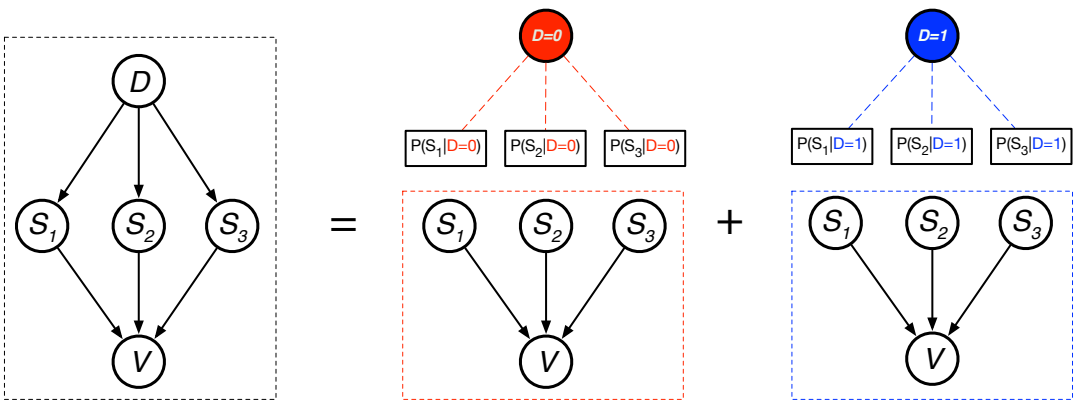


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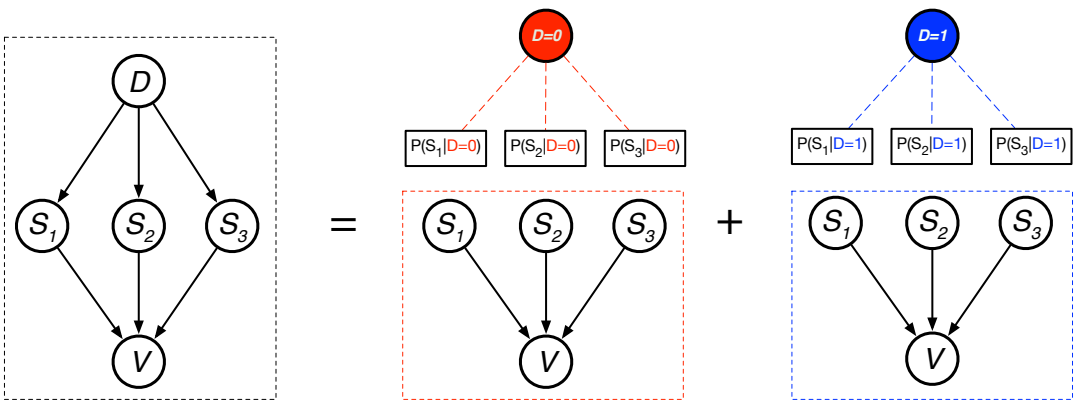


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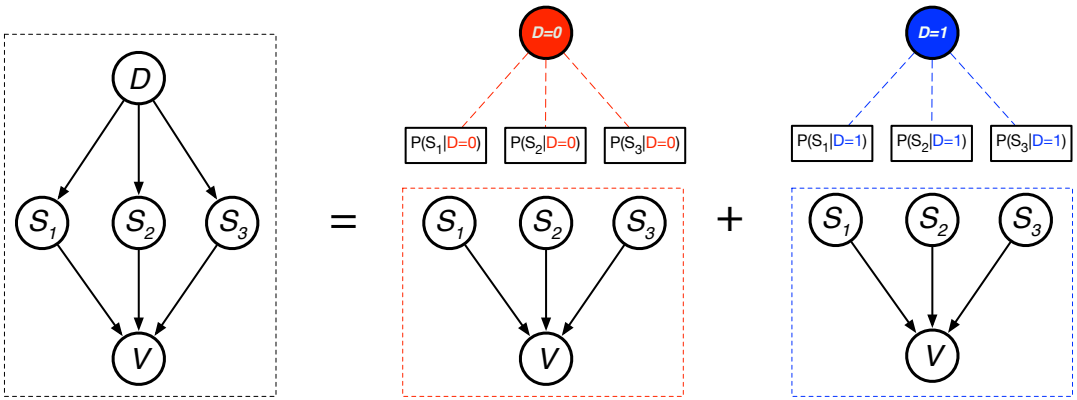
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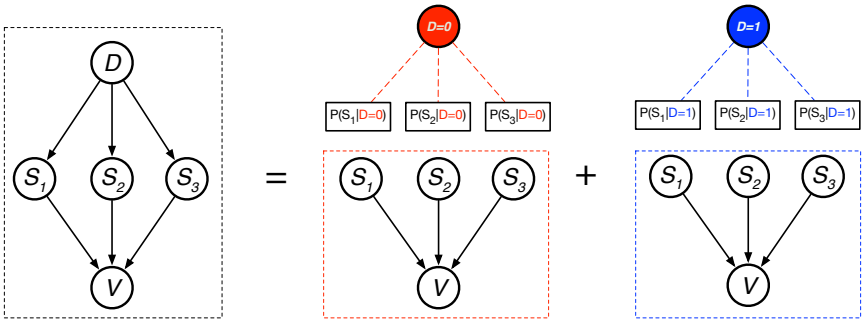
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  The set of instantiated nodes is called the **cutset**.
  By removing them, we cut the BN into polytrees

Worked example

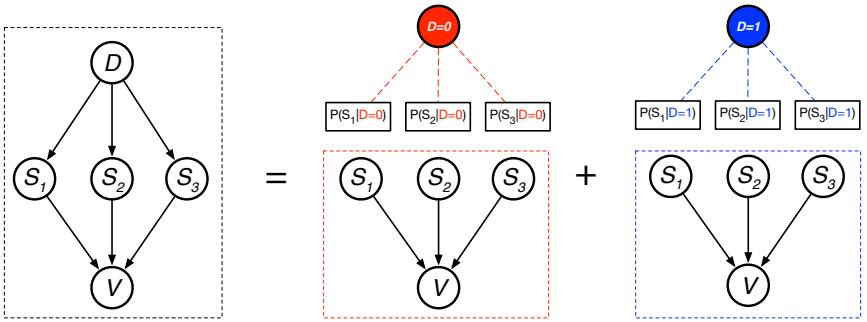
Worked example

How to calculate $P(V=1)$?



Worked example

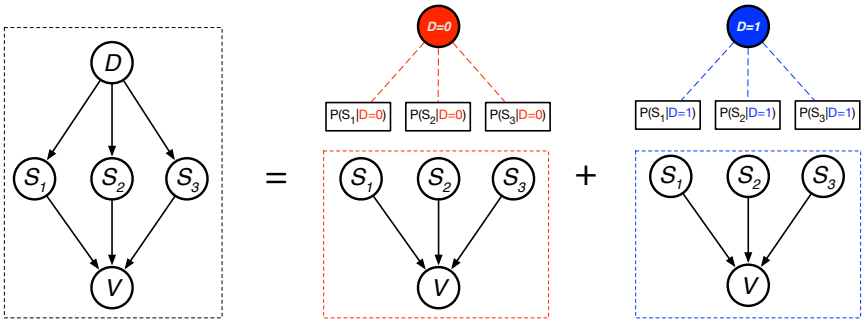
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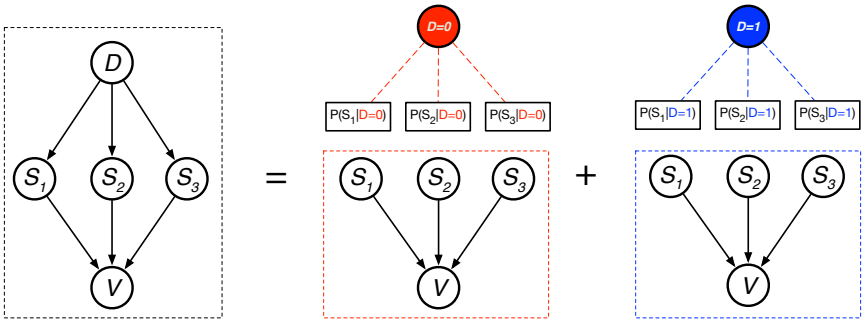


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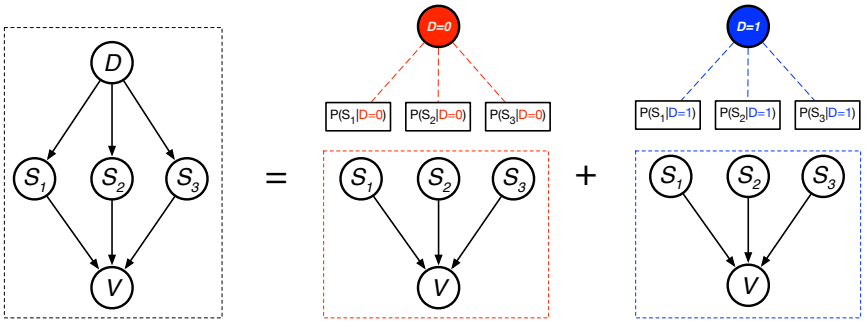


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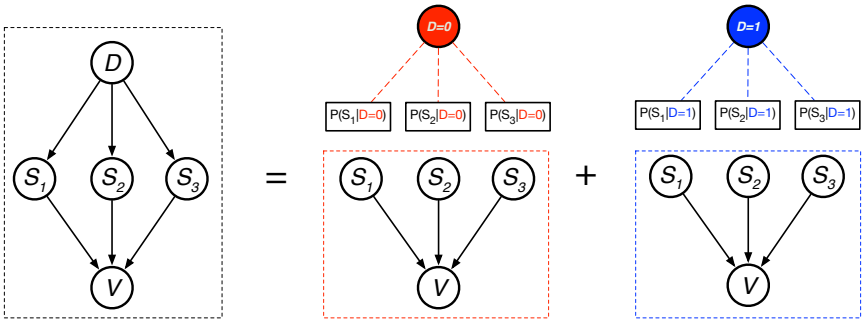
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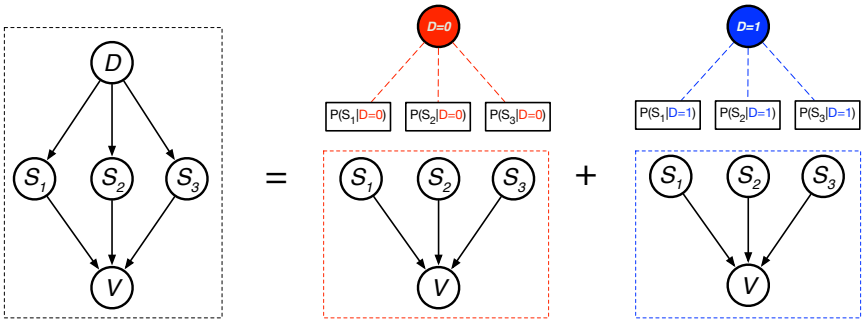
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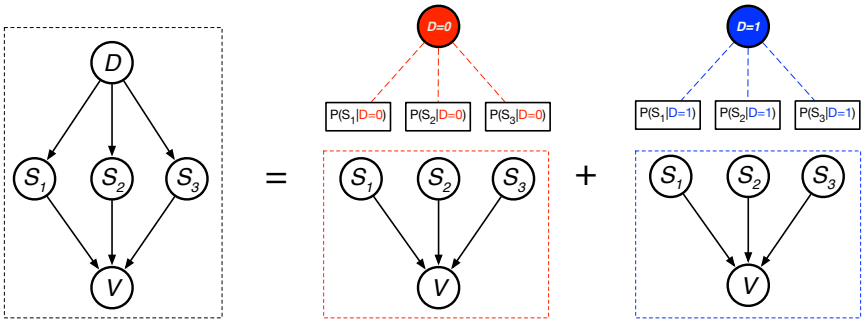
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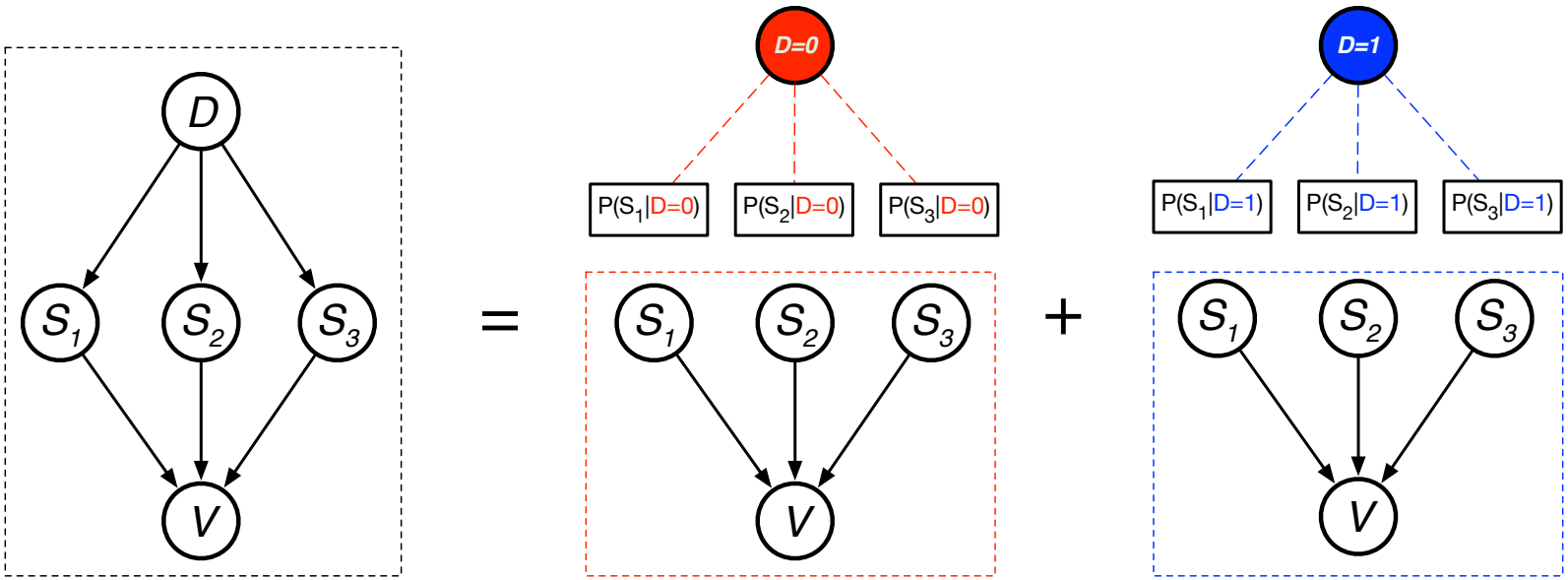
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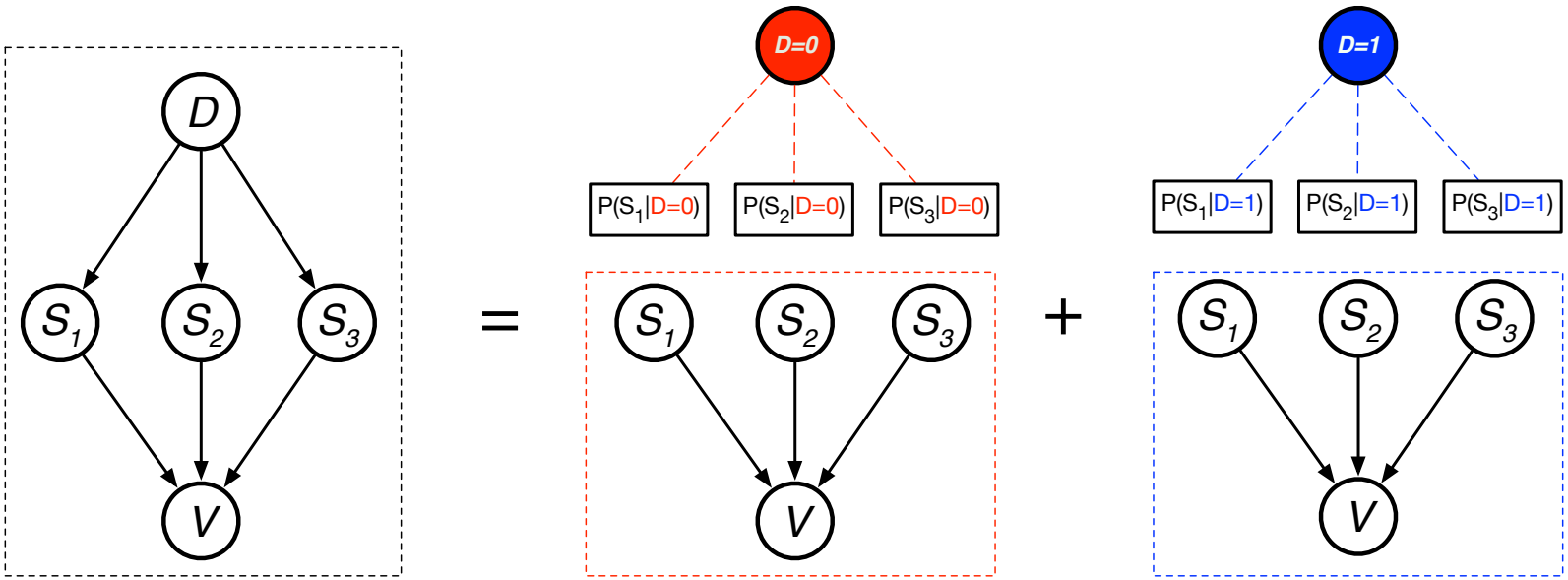
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Approximate inference in loopy BNs

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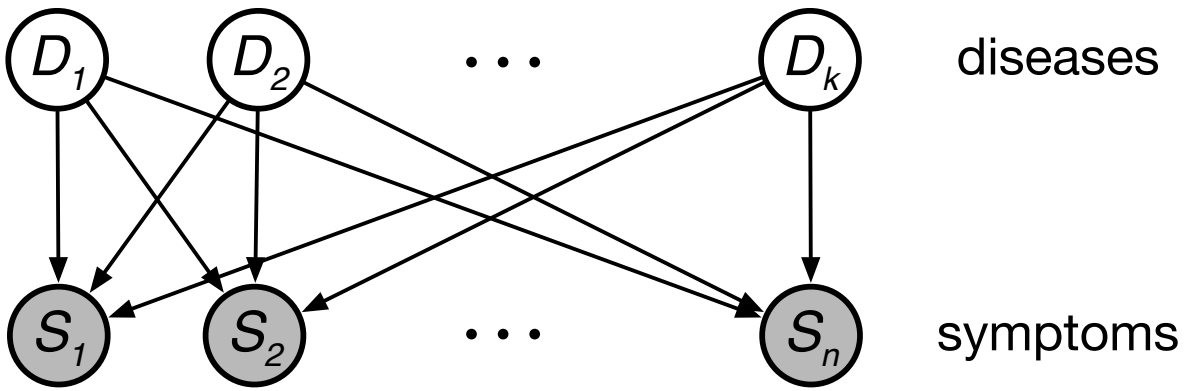
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2. Rejection sampling
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This lecture:

1. Prior sampling
2. Rejection sampling
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But before we can describe the above methods,
we must review some basic ideas in sampling ...

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- How can we estimate $P(H)$?

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Suppose we have a (biased) coin but we don't know the $P(H)$.

- How can we estimate $P(H)$?
- When will we get the correct probability?

Sampling

Basic Idea:

- Draw N samples from a sampling distribution S.

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We can apply similar intuition for inference in BNs -> Generate samples from a BN.

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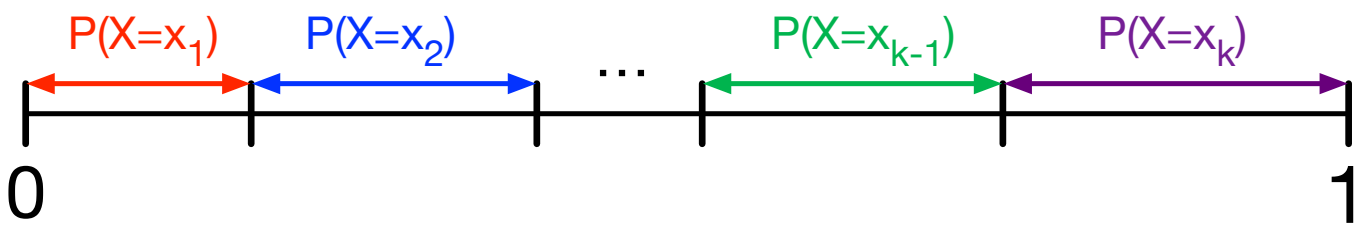
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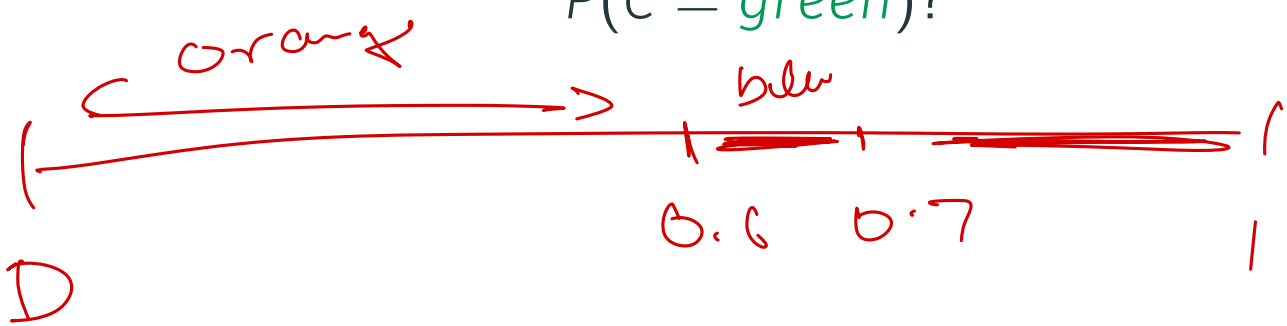
C	P(C)
orange	0.6
blue	0.1
green	0.3

1. Get sample u from uniform distribution over [0,1).
2. Assign discrete value of C.
   - $0 \leq u < 0.6, C \leftarrow orange$
   - $0.6 \leq u < 0.7, C \leftarrow blue$
   - $0.7 \leq u < 1, C \leftarrow green$

8 samples generated:

orange, green, orange, blue, orange, orange, green, orange

$P(C = green)$?

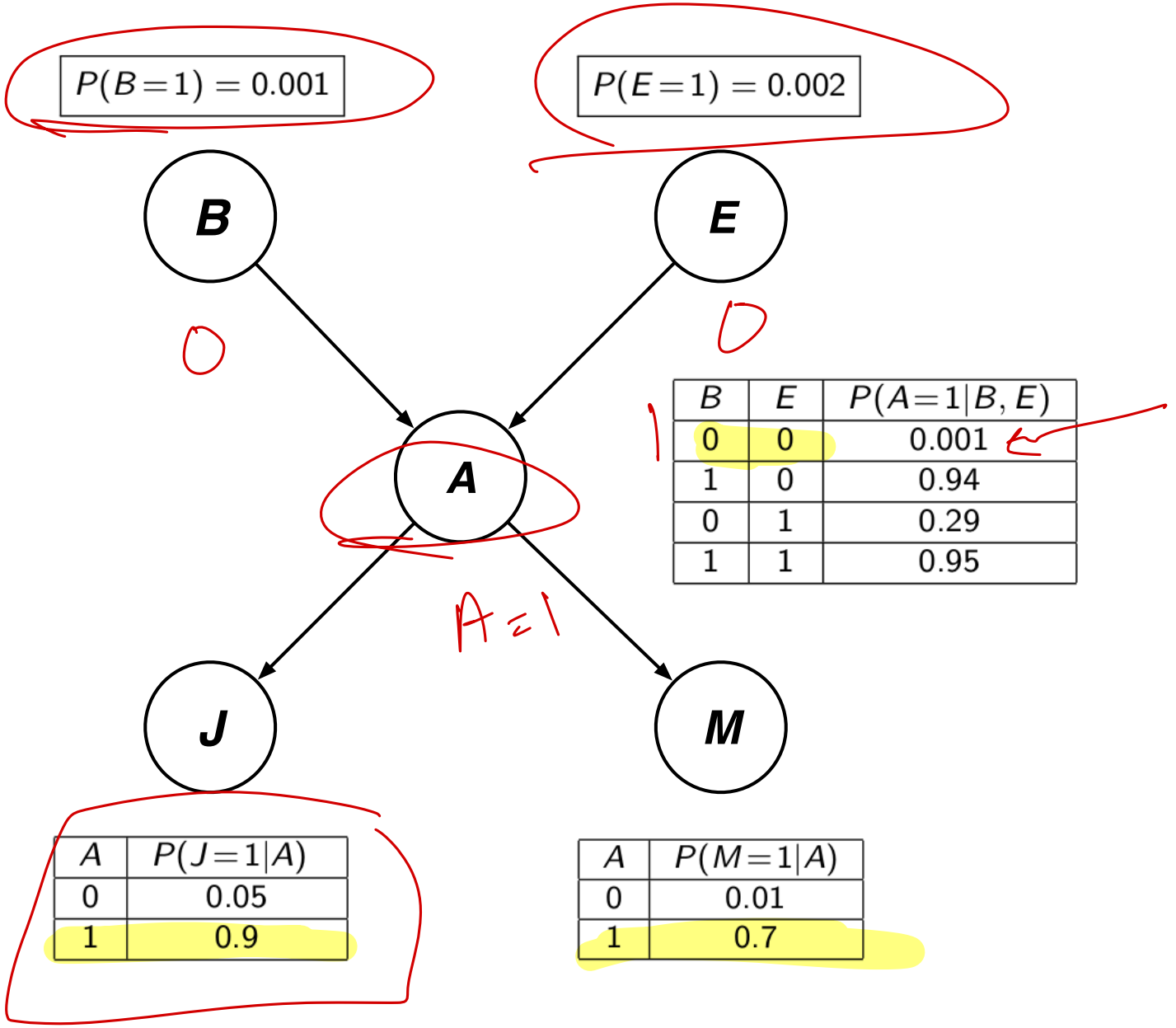


Generative Model

Belief Network is a generative model. We can sample from the distribution represented by the BN.

Idea: Generate one variable at a time in topological order.

Alarm example



B	E	$P(A=1\|B,E)$
0	0	0.001
1	0	0.94
0	1	0.29
1	1	0.95

A	$P(J=1\|A)$
0	0.05
1	0.9

A	$P(M=1\|A)$
0	0.01
1	0.7

Alarm example

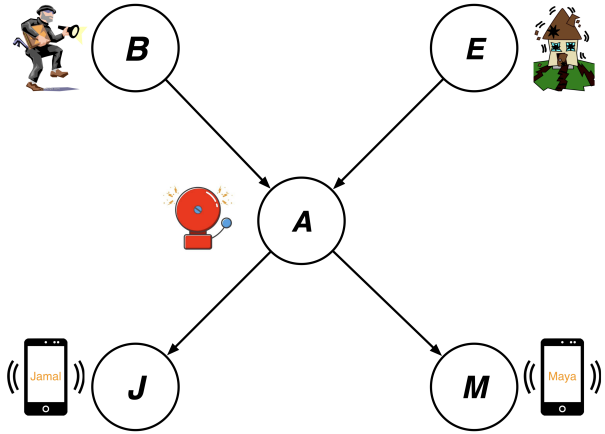
Alarm example

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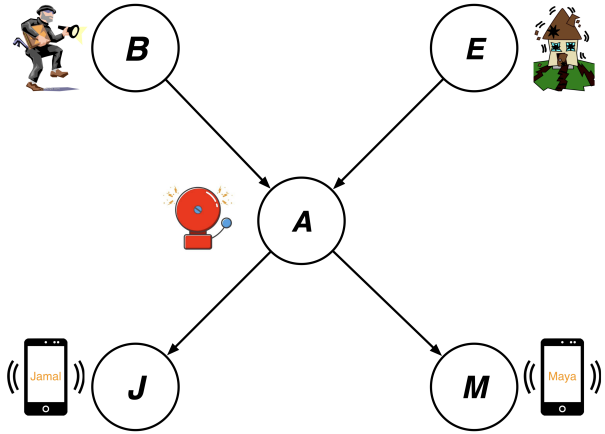


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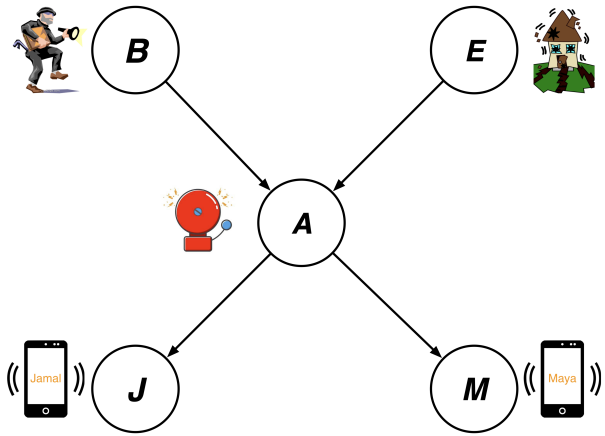
Alarm example

- **Joint sample**

  To draw a joint sample $\{b, e, a, j, m\}$ from $P(B, E, A, J, M)$, it is enough to draw the individual samples:

$$b \sim P(B)$$

$$e \sim P(E)$$

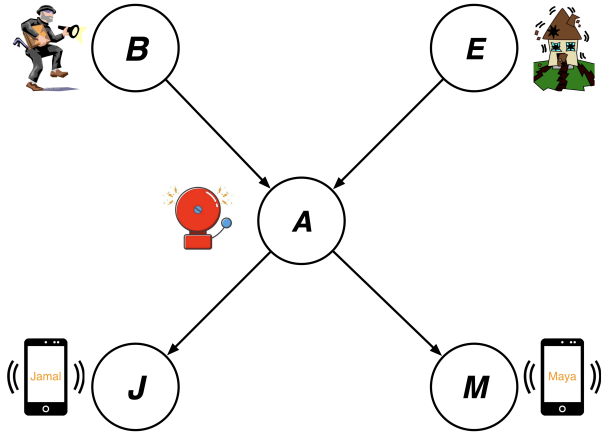


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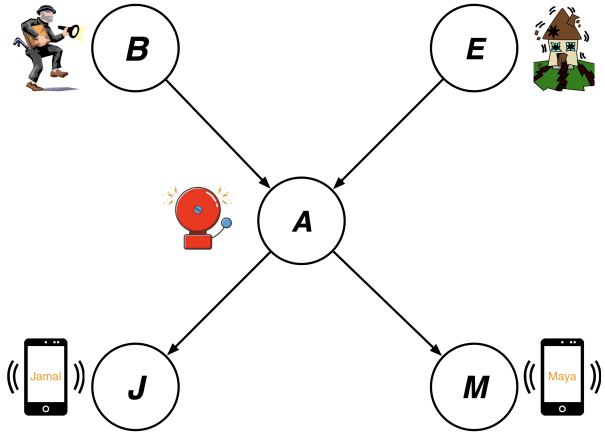


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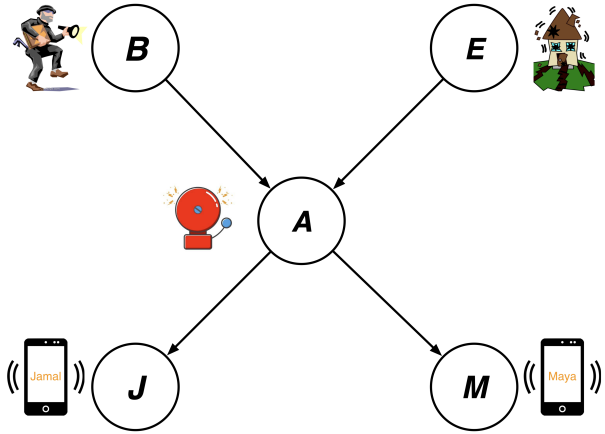


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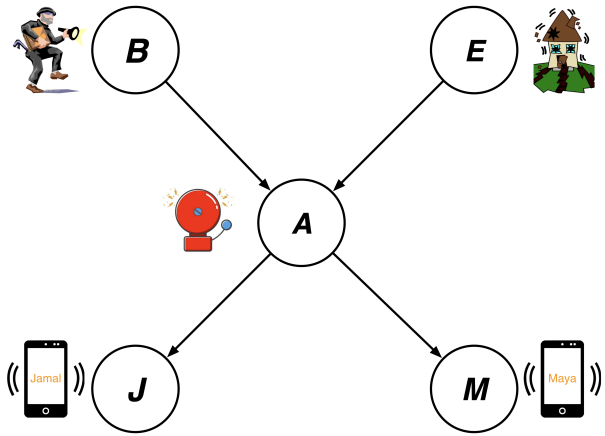


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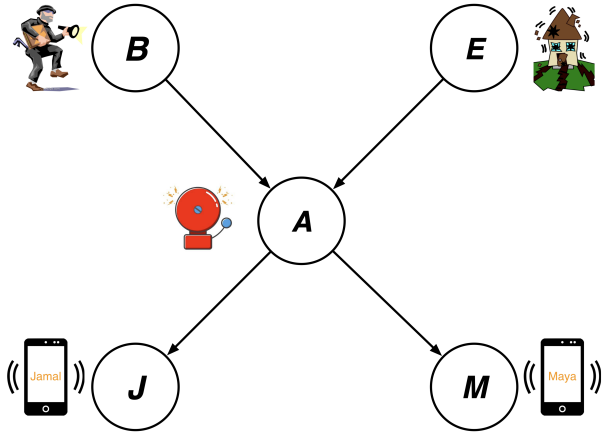
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- **Convergence in the limit**

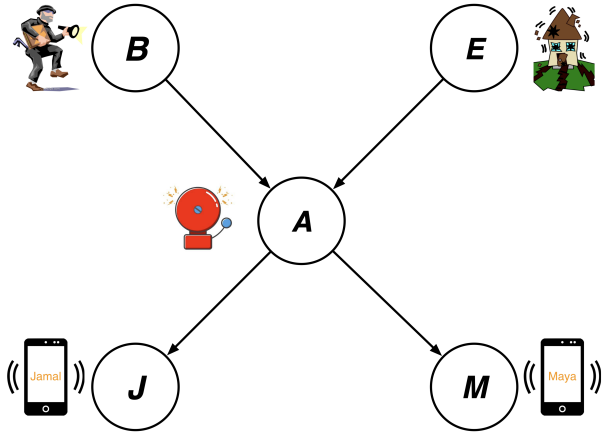
$$P(b, e, a, j, m) \; =$$

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$$



- **Convergence in the limit**

$$P(b, e, a, j, m) = \lim_{N \to \infty} \frac{\text{count}(B=b, E=e, A=a, J=j, M=m)}{N}.$$

Prior Sampling from the joint distribution in a discrete BN

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Let $\{X_1, X_2, \ldots, X_m\}$ be discrete random variables in a BN.

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- **Problem**

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  The BN defines a **generative model** with joint distribution

$$P(X_1, X_2, \ldots, X_m) = \prod_{i=1}^{m} P(X_i | \mathrm{pa}(X_i)).$$

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  To draw samples, we can simply take $X_i \sim P(X_i | \mathrm{pa}(X_i))$.

Approximate inference

Approximate inference

- Problem

Approximate inference

- **Problem**

  Let Q denote a set of query nodes.
  Let E denote a set of evidence nodes.

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- **Solutions**

  1. rejection sampling (very inefficient)
  2. likelihood weighting (more efficient)
  3. MCMC (most efficient)

Rejection sampling in BNs

Rejection sampling in BNs

- Problem

Rejection sampling in BNs

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  Let Q denote a set of query nodes.
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  How to estimate $P(Q = q | E = e)$?

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  Discard (or reject) the samples where $E \neq e$.

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- **Solution**

  Generate N samples from the BN's joint distribution.
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  Count the samples $N(q, e)$ where $Q=q$ and $E=e$.

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  Take the ratio of these counts:

Rejection sampling in BNs

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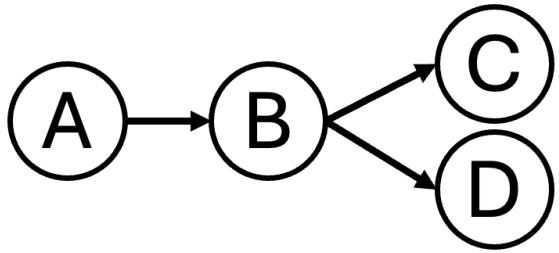
  Let Q denote a set of query nodes.
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- **Solution**

  Generate N samples from the BN's joint distribution.
  Discard (or reject) the samples where $E \neq e$.
  Count the samples $N(q,e)$ where $Q=q$ and $E=e$.
  Count the samples $N(e)$ where $E=e$.
  Take the ratio of these counts:

$$P(Q=q|E=e) \approx \frac{N(q,e)}{N(e)} \quad \text{where} \quad N(q,e) \leq N(e) \leq N$$

Example for rejection sampling



A	P(A)
a_0	1/5
a_1	4/5

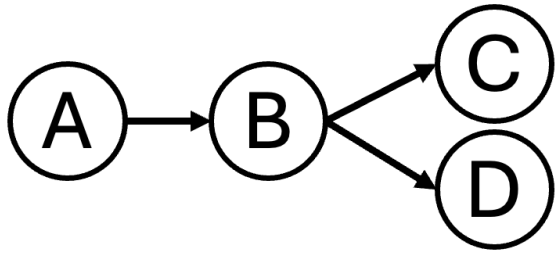
A	B	P(B|A)
a_0	b_0	1/4
a_0	b_1	3/4
a_1	b_0	1/3
a_1	b_1	2/3

B	C	P(C|B)
b_0	c_0	1/5
b_0	c_1	4/5
b_1	c_0	3/5
b_1	c_1	2/5

B	D	P(D|B)
b_0	d_0	3/4
b_0	d_1	1/4
b_1	d_0	1/3
b_1	d_1	2/3

Example for rejection sampling

Problem: Estimate $P(a_0|c_1, d_1)$



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a_0	1/5
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A	B	P(B|A)
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a_1	b_1	2/3

B	C	P(C|B)
b_0	c_0	1/5
b_0	c_1	4/5
b_1	c_0	3/5
b_1	c_1	2/5

B	D	P(D|B)
b_0	d_0	3/4
b_0	d_1	1/4
b_1	d_0	1/3
b_1	d_1	2/3

Example for rejection sampling

Problem: Estimate $P(a_0 | c_1, d_1)$

Samples:

a_1, b_1, c_0, d_0

a_0, b_1, c_1, d_0

a_1, b_1, c_0, d_1

a_1, b_1, c_1, d_1

a_1, b_0, c_0, d_0

a_0, b_1, c_1, d_1

N(e)

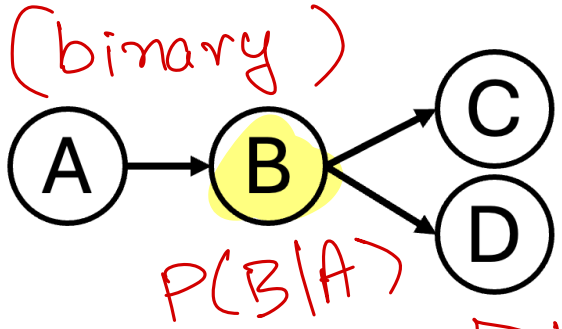
All the samples where E = e

N(a, e)

N(a = a₁, e)

2

1



A	P(A)
a_0	1/5
a_1	4/5

A	B	P(B|A)
a_0	b_0	1/4
a_0	b_1	3/4
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b_1	c_0	3/5
b_1	c_1	2/5

B	D	P(D|B)
b_0	d_0	3/4
b_0	d_1	1/4
b_1	d_0	1/3
b_1	d_1	2/3

Example for rejection sampling

Problem: Estimate $P(a_0|c_1, d_1)$

Samples:

a_1, b_1, c_0, d_0

a_0, b_1, c_1, d_0

a_1, b_1, c_0, d_1

a_1, b_1, c_1, d_1

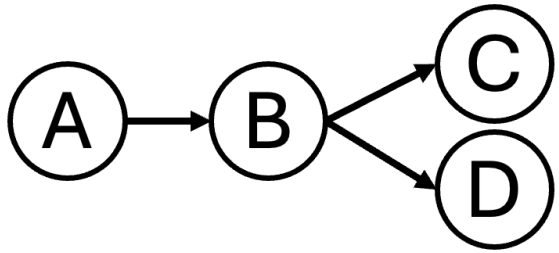
a_1, b_0, c_0, d_0

a_0, b_1, c_1, d_1

Q. How many samples will be rejected?

A. 6 **B.** 2 **C.** 4

D. 3 **E.** 0



A	P(A)
a_0	1/5
a_1	4/5

A	B	P(B|A)
a_0	b_0	1/4
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a_1	b_0	1/3
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B	C	P(C|B)
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b_0	c_1	4/5
b_1	c_0	3/5
b_1	c_1	2/5

B	D	P(D|B)
b_0	d_0	3/4
b_0	d_1	1/4
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b_1	d_1	2/3

Example for rejection sampling

Problem: Estimate $P(a_0|c_1, d_1)$

Samples:

~~a_1, b_1, c_0, d_0~~

~~a_0, b_1, c_1, d_0~~

~~a_1, b_1, c_0, d_1~~

a_1, b_1, c_1, d_1

~~a_1, b_0, c_0, d_0~~

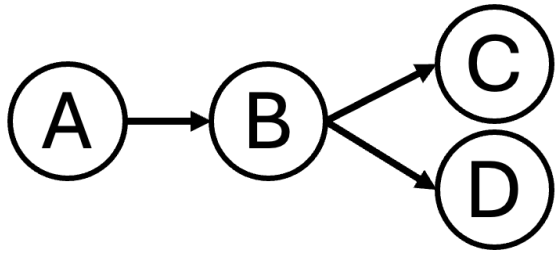
a_0, b_1, c_1, d_1

count $(Q = q, E = e)$

1/2

count $(E = e)$

= 2



A	P(A)
a_0	1/5
a_1	4/5

A	B	P(B|A)
a_0	b_0	1/4
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b_0	c_1	4/5
b_1	c_0	3/5
b_1	c_1	2/5

B	D	P(D|B)
b_0	d_0	3/4
b_0	d_1	1/4
b_1	d_0	1/3
b_1	d_1	2/3

Q. Estimate of $P(a_0|c_1, d_1)$ using rejection sampling?

A. 1/2 **B.** 2/3 **C.** 1/4

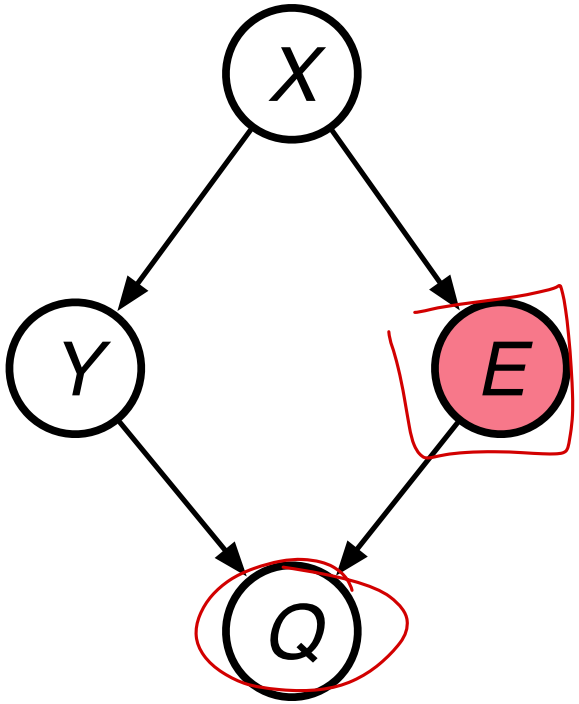
D. 1/3 **E.** 0

Example for rejection sampling

Example for rejection sampling

- Sample N times:

$$
\begin{aligned}
x_i &\sim P(X) \\
y_i &\sim P(Y|X=x_i) \\
e_i &\sim P(E|X=x_i) \\
q_i &\sim P(Q|Y=y_i, E=e_i)
\end{aligned}
$$

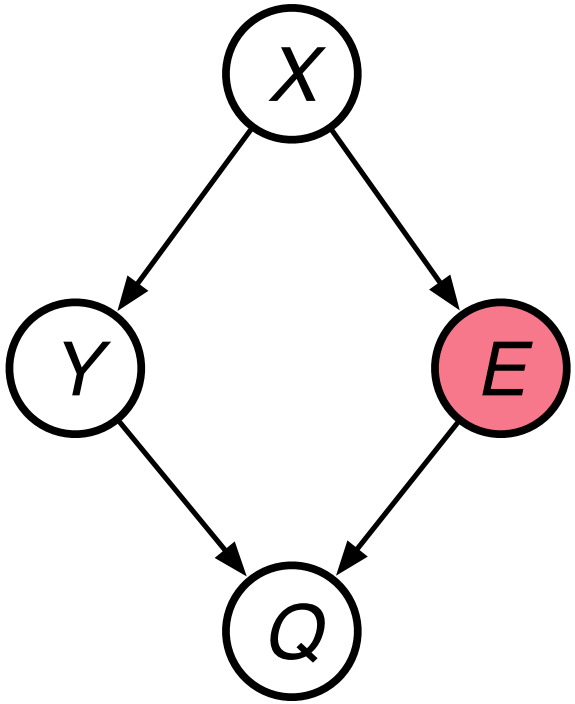


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- Define the indicator function:



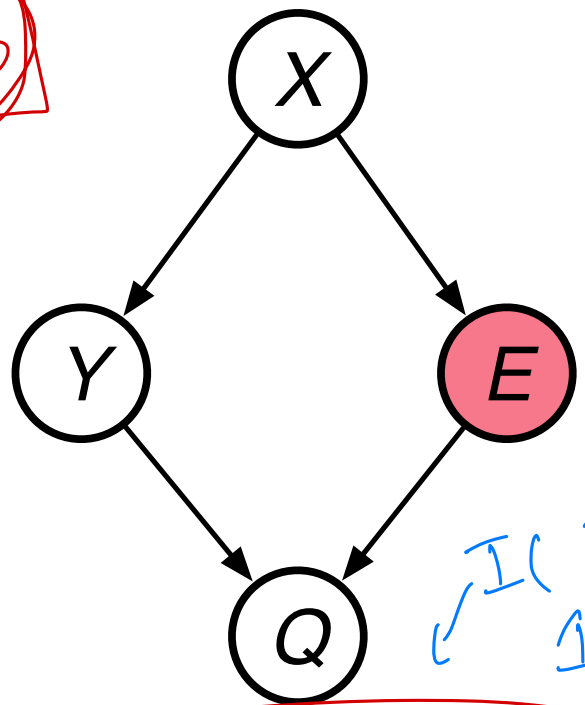
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- Define the indicator function:

$$
I(z, z') = \begin{cases} 1 & \text{if } z = z' \\ 0 & \text{if } z \neq z' \end{cases}
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Example for rejection sampling

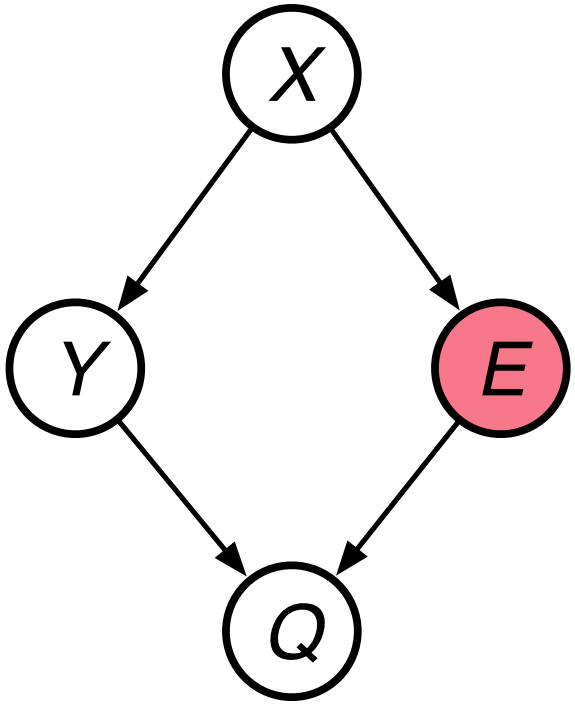
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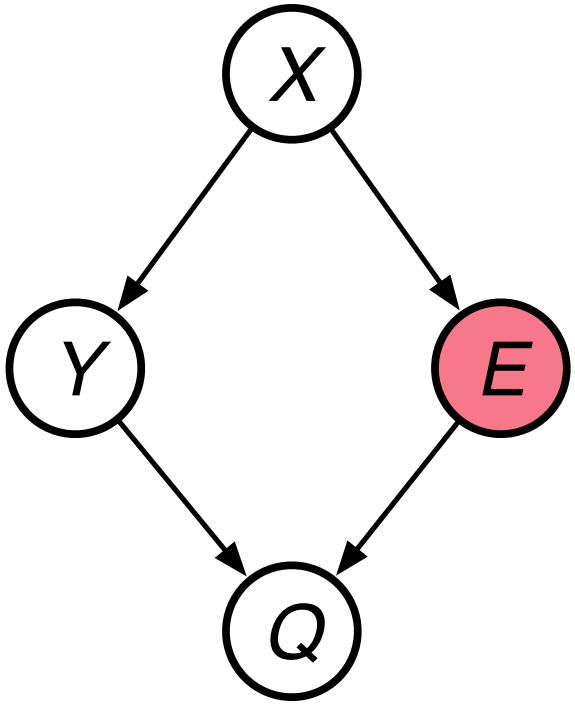
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Example for rejection sampling

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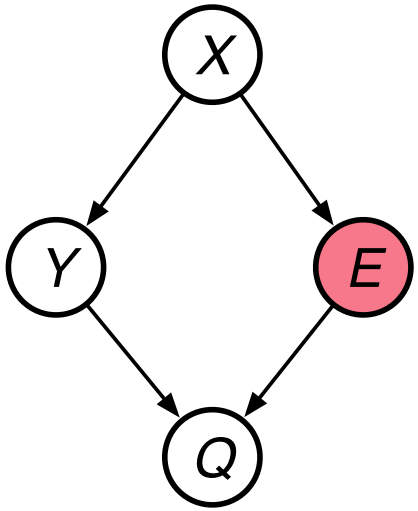
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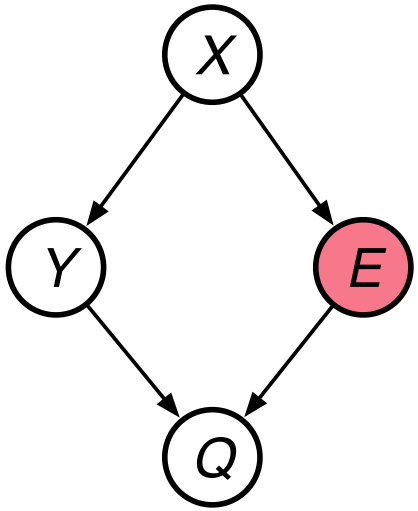
$$
P(Q=q|E=e) \approx \frac{N(q,e)}{N(e)} = \frac{\sum_{i=1}^{N} I(q, q_i) I(e, e_i)}{\sum_{i=1}^{N} I(e, e_i)}
$$

Properties of rejection sampling



Properties of rejection sampling

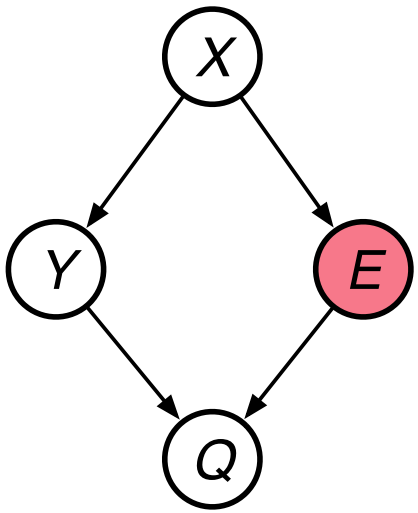
- It converges in the limit:



Properties of rejection sampling

- It converges in the limit:

$$\lim_{N \to \infty} \frac{N(q, e)}{N(e)} = P(Q = q | E = e)$$

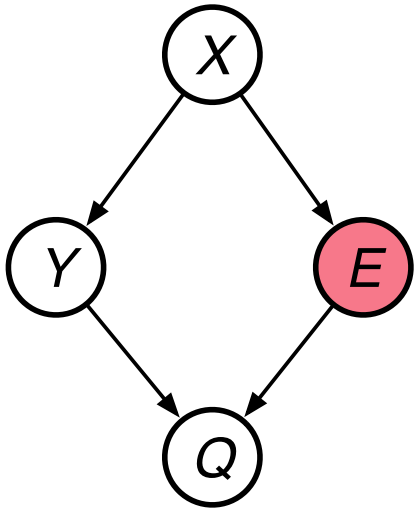


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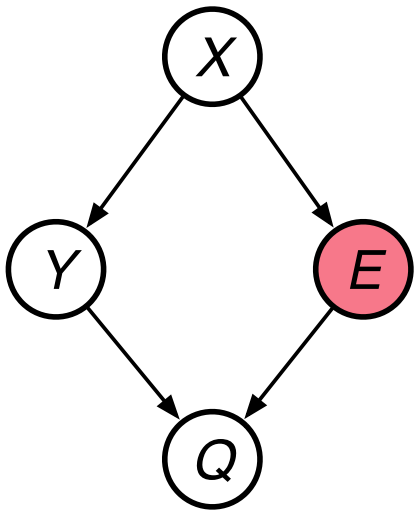
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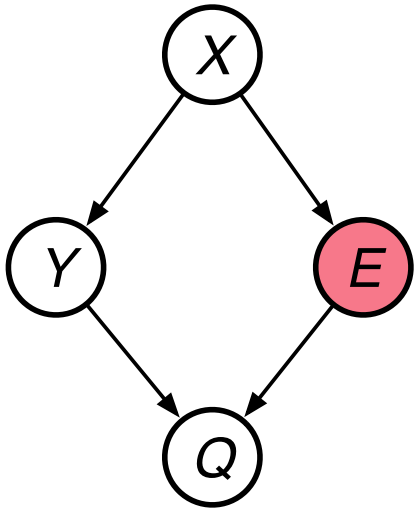
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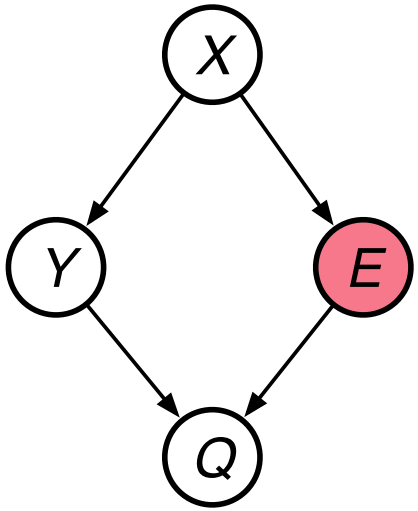
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How can we do better?

2. Likelihood weighting

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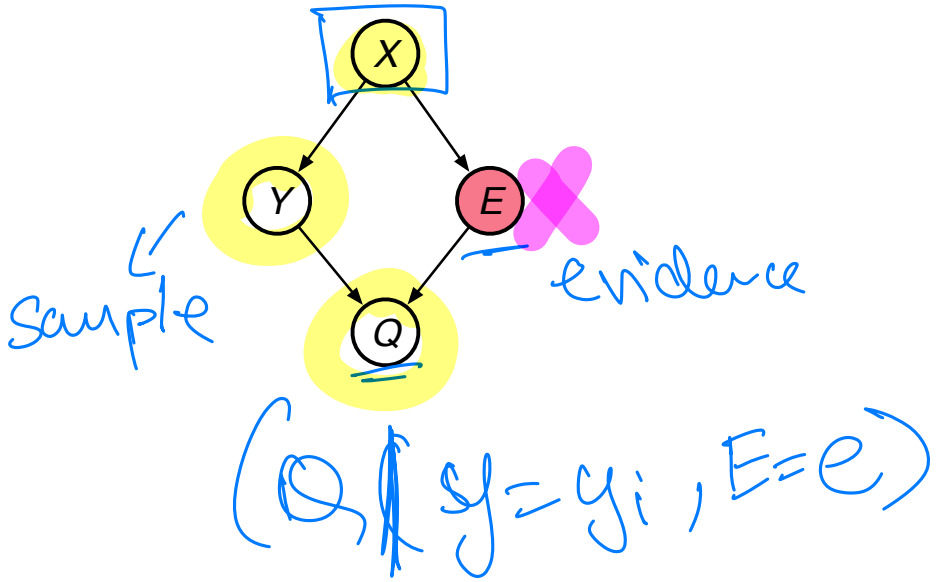
- **Benefits**

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  No wasted computation.
  Faster convergence. than rejection sampling.

Example for likelihood weighting

Example for likelihood weighting

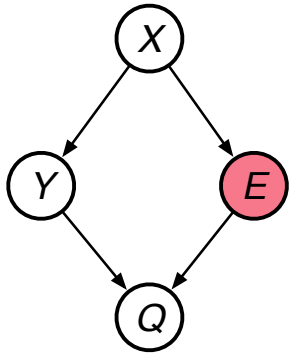
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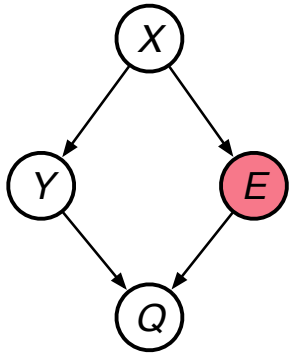


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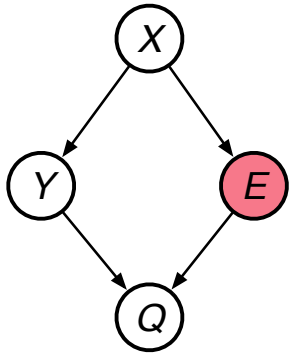


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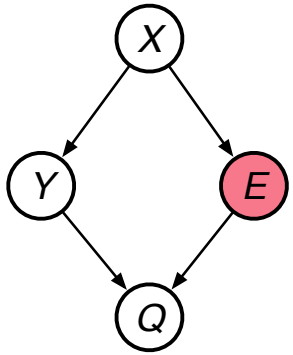


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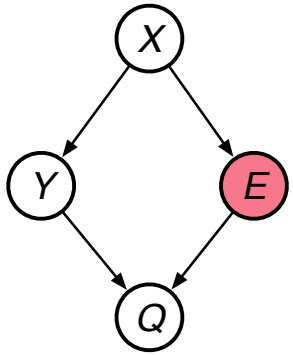
we are not sampling E → because we set it to e

Example for likelihood weighting

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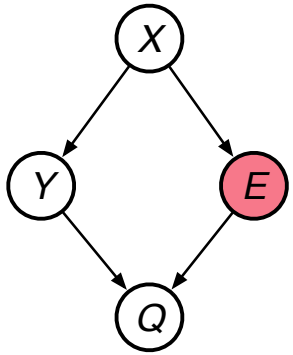
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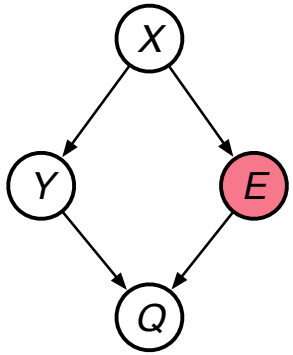
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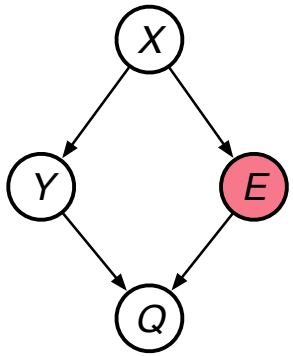
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counting

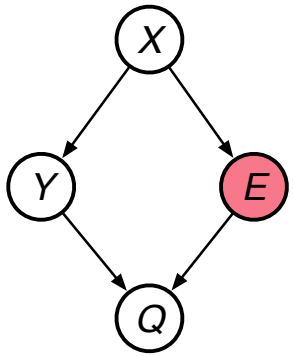
likelihood weight

instead of counting we use the CPT of E

$I(\quad)$

Example for likelihood weighting

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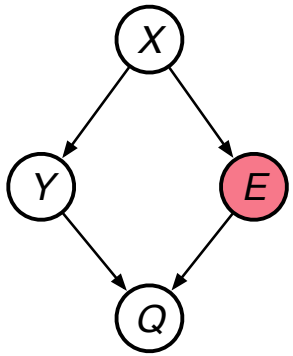
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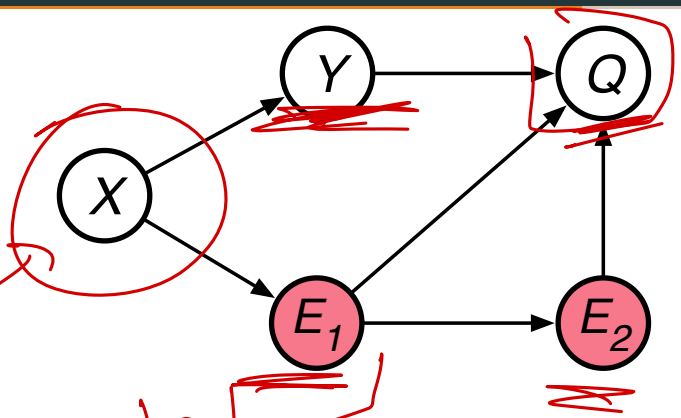
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Example with multiple evidence nodes

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How to estimate $P(Q = q | E_1 = e, E_2 = e')$?



where do i sample X from?

P( )?

Y?

P(X)

$P(y | X = x_i)$

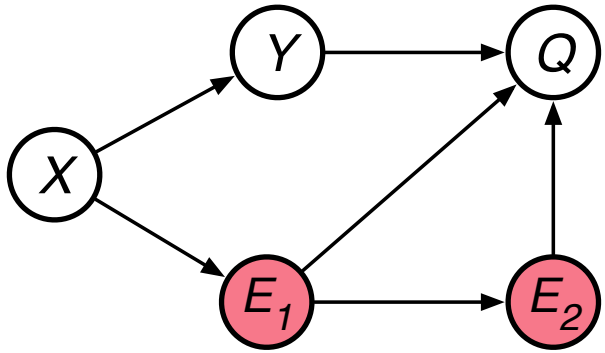
$\theta \rightarrow$

$P(\theta | y = y_i, e_1, e_2)$

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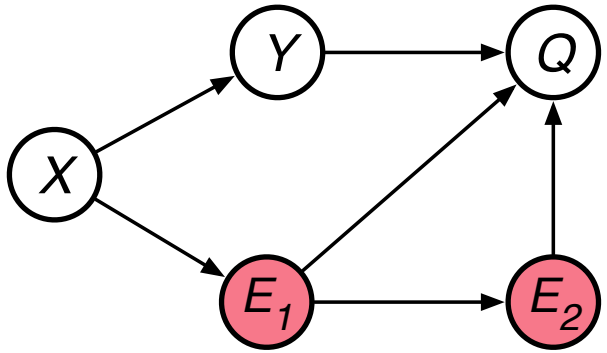


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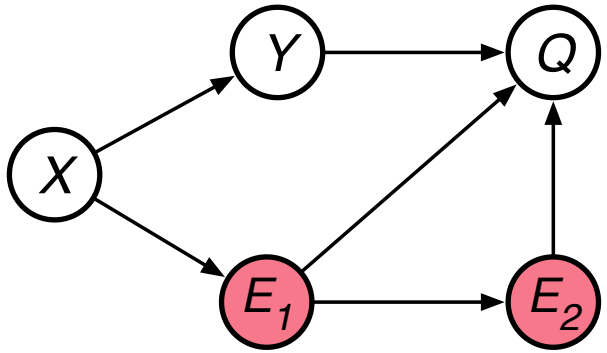


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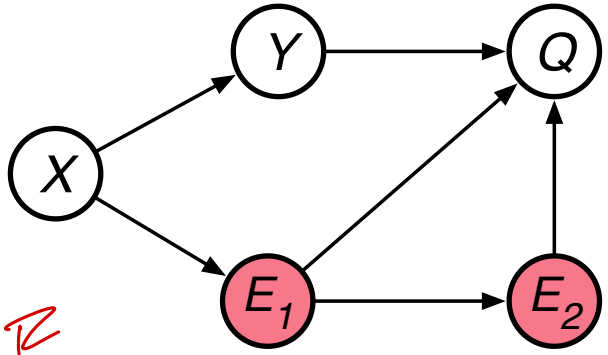


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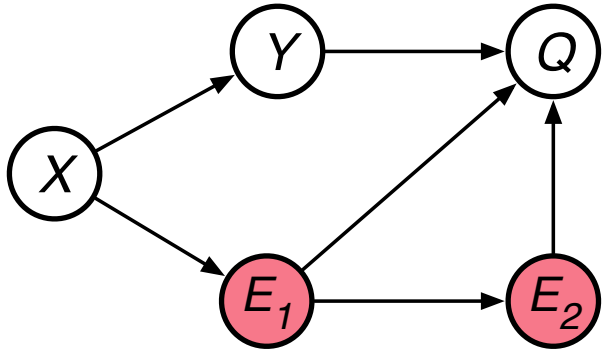
Don't sample E we set to to

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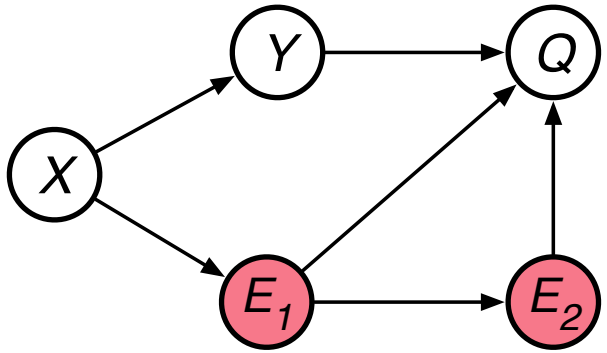
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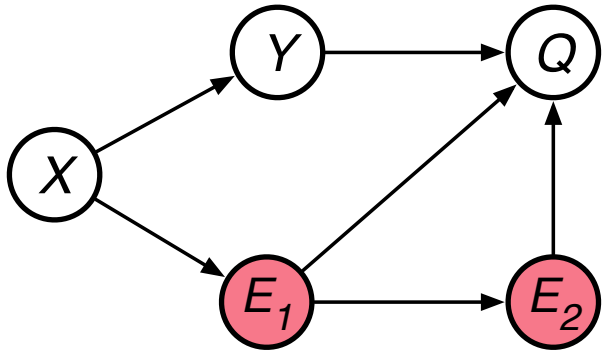
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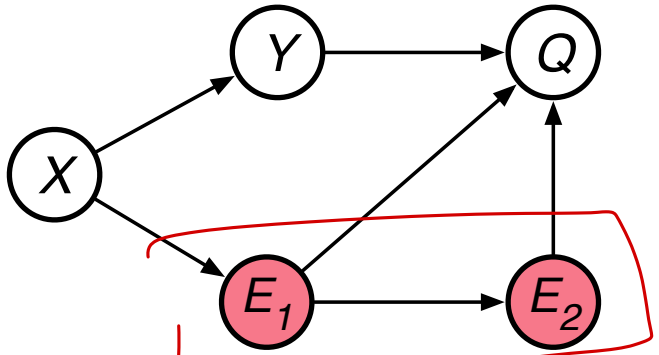
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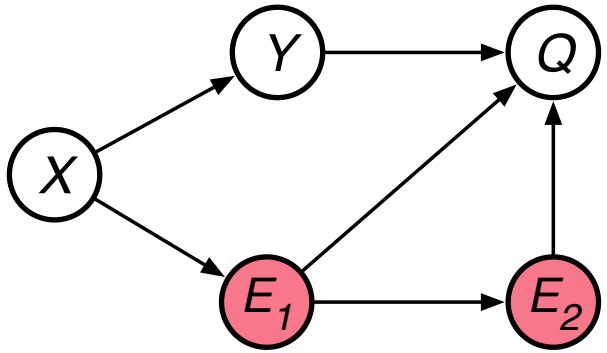
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product of likelihood weights

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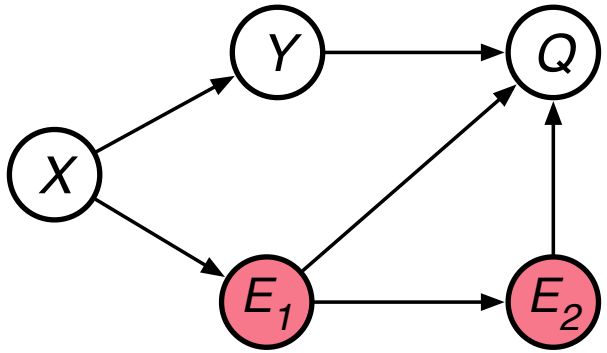
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- Compare to rejection sampling:

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Example for likelihood weighting sampling

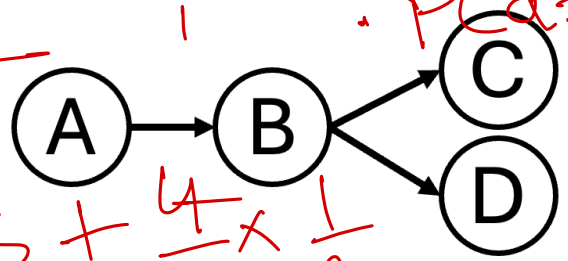
Problem: Estimate $P(a_0|c_1, d_1)$

Samples:

a_0, b_1, c_1, d_1

a_1, b_0, c_1, d_1

a_0, b_1, c_1, d_1



A	P(A)
a_0	1/5
a_1	4/5

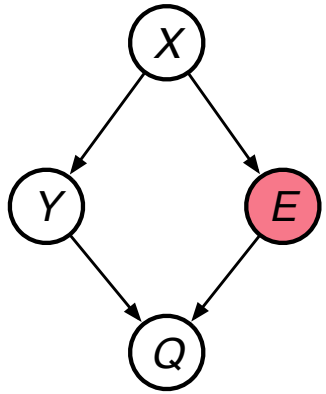
A	B	P(B|A)
a_0	b_0	1/4
a_0	b_1	3/4
a_1	b_0	1/3
a_1	b_1	2/3

B	C	P(C|B)
b_0	c_0	1/5
b_0	c_1	4/5
b_1	c_0	3/5
b_1	c_1	2/5

B	D	P(D|B)
b_0	d_0	3/4
b_0	d_1	1/4
b_1	d_0	1/3
b_1	d_1	2/3

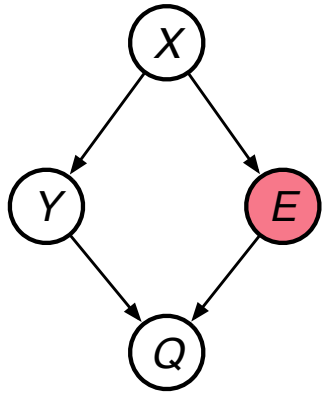
Q. Estimate of $P(a_0|c_1, d_1)$ using likelihood weighting?

Properties of likelihood weighting



Properties of likelihood weighting

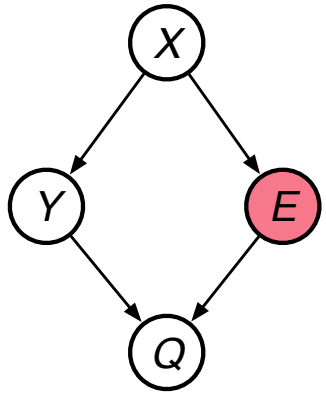
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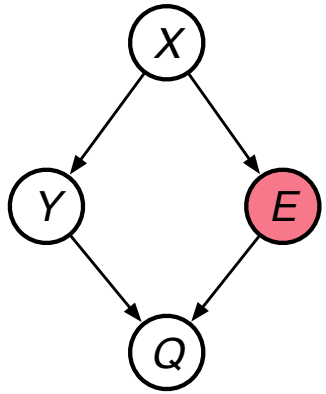
$$\lim_{N\to\infty} \frac{\sum_{i=1}^{N} l(q, q_i)\, P(E=e|X=x_i)}{\sum_{i=1}^{N} P(E=e|X=x_i)} =$$



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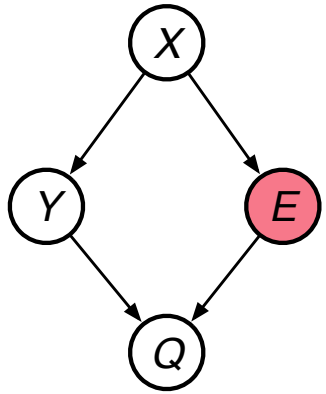
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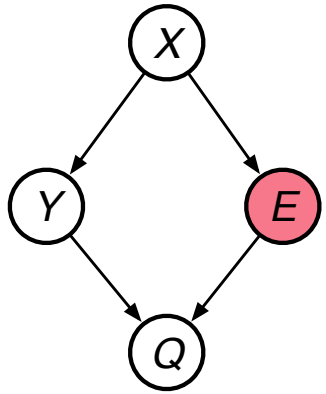


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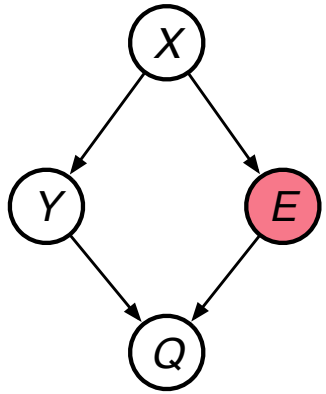
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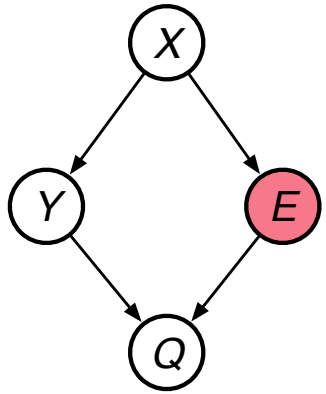
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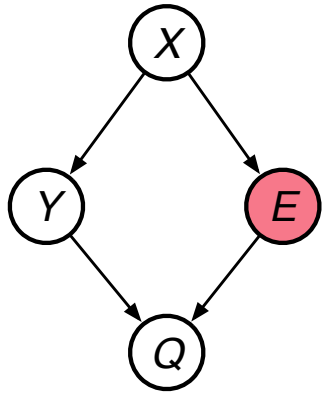
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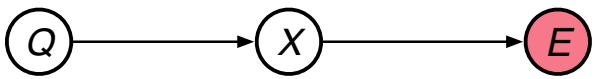
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The worst case for likelihood weighting is when rare evidence is descended from query nodes.

That’s all folks!